Safety Stock for Machine Error: Standby Labour, Screening Depth and Correlated Risk in LLM Services

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Abstract

When a firm puts a large language model (LLM) into service, it takes on a new stochastic cost: the errors that people have to absorb. We study how the firm should size a standby human workforce when automated guardrail layers screen the errors first. Standby capacity is a newsvendor quantile of the daily count of human-handled incidents, and a 95% service level is optimal only when an unhandled incident costs twenty times a unit of capacity. Over-dispersion pushes the safety stock above what the dispersion index suggests, and stacking screens under independence overstates diversification when the layers fail on the same inputs. We show that the multiplicative guardrail formula is the independence case of sharp dependence bounds, that a weak correlation between high-clearance layers inflates residual risk by tens of per cent, and that under a common-difficulty model residual risk decays only polynomially in the number of layers. With integer capacity and a staffing floor, the cost-minimising depth of screening is in general not unimodal. We evaluate the model on measured outputs of three small open models passed through three guardrail layers, using simulated arrivals and public traces. A task-specific verifier cut required capacity by 78 to 96% but rejected 15 to 75% of correct answers, the share of hallucinations passing two layers was 1.2 to 3.3 times the independence prediction, rising with the clearance of the layers, and ignoring day-to-day variation raised expected cost by 9 to 65%. Because arrivals are simulated, the numbers illustrate the method.

Keywords: LLM operations, hallucination, safety stock, newsvendor, factor substitution, correlated risk, over-dispersion

1. Introduction

A firm that deploys a large language model (LLM) to answer questions automates a task, but it also creates a new kind of random event: an answer that is wrong, or not supported by the evidence, which someone must then find and fix. A support assistant that makes up a refund policy, or a drafting tool that cites a court decision that was never handed down, turns the firm's own production process into a source of incidents. The resulting problem will be familiar to

economists from inventory theory and from the theory of production under uncertainty. To minimise expected cost, the firm has to choose an automated screening technology, which behaves like capital (scalable but imperfect), and a standby human workforce, which behaves like labour (flexible but expensive), to absorb whatever the screening lets through. How large should the standby workforce be? How much screening is worth buying? And how far can one substitute for the other?

Two rules of thumb are widely used, and in economic terms both are mistaken. The first is certainty-equivalent staffing, where the team is sized for the mean incident load as though uncertainty about that load were free. Even if incident counts are exactly Poisson, a team sized this way is short on between a quarter and a half of days once the rate reaches one event per day (Section 3.1); in our replays such a plan was exceeded on 28 to 49% of days. The second is what we call the diversification fallacy. Guardrail layers are stacked and each is credited with its stand-alone effectiveness, so that three layers catching 63% of hallucinations apiece are taken to leave about 5%. This arithmetic is valid only if the layers fail independently. When they tend to fail on the same inputs, much as correlated assets do in a portfolio, the value of redundancy is overstated and residual risk is understated.

The paper addresses three questions. First, what size of standby workforce minimises cost, and what does a given service-level target imply about the cost of an unhandled incident? Second, how does the optimal amount of automated screening depend on the price of human labour relative to compute, and how much of its apparent benefit remains once we account for dependence between layers and for their costs in latency, false blocks and failures? Third, do the stochastic assumptions behind these answers, namely Poisson arrivals with at most modest over-dispersion, hold up on public data from comparable real systems?

To answer them we combine a formal model with an empirical evaluation. The model (Section 3) treats the daily count of human-handled incidents as a random variable and sets standby capacity at a newsvendor-type quantile. It then shows how over-dispersion, thinning by screening and dependence between layers move that quantile and the cost that goes with it. The evaluation (Sections 4 and 5) draws on the measured outputs of three small open models, each of which answered the same 300 SQuAD 2.0 questions (and, in a held-out replication, 1,000 further questions), passed through three guardrail layers that were actually executed. Arrival times are simulated. Their day-to-day variability is informed by public traces from an LLM service, from

LLM providers' incident reports, from company ticket systems and from court decisions involving fabricated citations. Apart from the court-case database, which does so only in part, none of these traces records hallucinations, and we are not aware of any public operator log that does.

Three findings stand out. First, a single task-specific verifier delivered most of what the cascade achieved. It reduced required capacity by 78 to 96%, but it also blocked 15 to 75% of correct answers, so much of the saving was simply a shift of cost from standby labour to false rejections. A routed LLM judge did not demonstrably pay for itself under a two-second fail-open timeout, although larger judges from other families did so modestly in the replication. Second, the share of hallucinations that got past two layers was 1.3 to 1.6 times the independence prediction in the first sample. In a held-out replication on 1,000 fresh questions it was 1.2 to 1.3 times with the same local judge and 1.6 to 3.3 times with larger judges from other families, with intervals that exclude one, and it rose when routing kept some answers from the second layer. Third, mild day-to-day variation in demand, with a coefficient of variation of 0.3 to 0.44, was enough for a Poisson-sized plan to be exceeded on 23 to 32% of days without guardrails and on 7 to 16% of days with the best cascade, compared with a target of 5%. Measured in cost, ignoring this variation raises expected daily cost by 9.4% to 65.3% across the cases we examine (Table 6). Real request arrivals from a public chatbot log show a daily-rate coefficient of variation of about 0.27, in line with the lower scenario value.

Our contribution lies mainly in application and measurement, and we try to be explicit about which results are already known. The newsvendor solution and the negative-binomial treatment of over-dispersed demand are classical (Arrow et al., 1951; Porteus, 2002; Cameron and Trivedi, 1998), and so is the Poisson quantile rule for staffing (Whitt, 2007). Staffing under uncertain arrival rates (Whitt, 2006; Bassamboo et al., 2010) and dependent failures in redundant systems (Eckhardt and Lee, 1985; Littlewood and Miller, 1989) are likewise established topics. We apply these tools to a new input, standby labour for machine errors, and add three results. We show how thinning by screening interacts with over-dispersion (Proposition 3). We recast the multiplicative guardrail formula as the independence special case of a set of sharp dependence bounds, give an exact link between the correlation of two layers and the inflation of residual risk, derive a closed form under a common-difficulty model in which residual risk decays only polynomially in depth, and measure the dependence for hallucination-grounding layers

(Proposition 4). We show that when capacity is integer-valued and standing staff has a floor, cost is in general not unimodal in the depth of screening, reduce the search to the breakpoints of the capacity schedule, show that any depth beyond one layer is justified only by an assumption about dependence, and derive comparative statics in the ratio of labour to compute prices and in dependence (Proposition 5). Proposition 6 converts the service-level target into the shortfall cost that would justify it and puts a price on misspecifying dispersion and dependence. Two natural extensions, a typed model of heterogeneous failure classes and a model of reviewers' attention as a moral-hazard problem, are left to future work (Section 6.3).

The paper sits in the operations research tradition of stochastic inventory and workforce-capacity models. Its contribution is a planning method, grounded in a formal model and evaluated on measured system outputs, in which known solutions are carried into a new problem domain, namely standby labour for machine errors. The evaluation has clear limits, which we discuss in Section 6.3: arrivals are simulated, the models are small and the task is extractive question answering. The numbers are there to illustrate the method, and it is the method and its diagnostics that we expect to transfer. The rest of the paper is organised as follows. Section 2 reviews related work and Section 3 develops the model. Sections 4 and 5 describe the evaluation and report its results, Section 6 discusses implications and limitations, and Section 7 concludes.

2. Background and Related Work

2.1 Safety stock and staffing under random demand

Choosing capacity in the face of random demand is the oldest problem in stochastic inventory theory. Arrow et al. (1951) showed that the optimal stock is a quantile of the demand distribution, with the quantile determined by the ratio of shortage to holding costs; Porteus (2002) provides the modern treatment. Gallego and Moon (1993) review distribution-free versions of the newsvendor rule, which use only the mean and variance of demand, and Van Mieghem (2003) reviews capacity management and investment under demand uncertainty.

Service operations apply the same reasoning to staffing. The square-root safety-staffing rule goes back to Halfin and Whitt (1981) and underlies the dimensioning of large call centres (Borst et al., 2004); Gans et al. (2003) and Aksin et al. (2007) review the field, Green et al. (2007) adjust staffing for demand that varies over time, and Whitt (2007) sets out the Poisson-quantile

rule that we use as our baseline. Harrison and Zeevi (2005) develop a fluid-model approach to staffing.

A long line of work asks what happens when the arrival rate is itself uncertain. Empirical work on call centres has found that arrivals are Poisson only conditional on a random, day-specific rate, which makes daily counts over-dispersed (Brown et al., 2005), and Avramidis et al. (2004) propose doubly stochastic models for daily arrivals. Jongbloed and Koole (2001) propose Poisson mixtures to manage this uncertainty, and Whitt (2006), Steckley et al. (2009) and Bassamboo et al. (2010) study staffing and performance under uncertain arrival rates and forecast errors, with Bassamboo et al. (2010) showing that safety-staffing principles have to be revisited. Count-data econometrics offers the negative binomial as a remedy (Cameron and Trivedi, 1998). We bring these tools to a stream whose arrivals are produced by the firm's own technology, add automated screening as a second, substitutable input, and check whether the usual caveats still apply.

2.2 Managing AI and human oversight

Berente et al. (2021) argue that AI raises managerial problems that earlier information technologies did not, owing to its autonomy, its capacity to learn and its inscrutability. A related literature looks at how organisations split work between people and AI and identifies a tension between automating tasks and augmenting human judgement (Raisch and Krakowski, 2021). In both, human review of AI output plays a central role. Yet neither says much about how large the human reserve needs to be, a question about quantities that depends on how errors arrive.

2.3 Hallucination and its mitigation

Hallucination, meaning the generation of content that is wrong or unsupported by the source, is well documented as a pervasive feature of neural text generation (Ji et al., 2023). Approaches to mitigation include retrieval grounding (Lewis et al., 2020), consistency checks based on repeated sampling (Manakul et al., 2023), safeguard classifiers (Inan et al., 2023) and verification by an LLM acting as a judge (Zheng et al., 2023). This work assesses detectors by their accuracy. It does not ask what that accuracy is worth in human capacity and in money, and that translation is what we attempt here.

2.4 Queueing and counting-process models of AI work

Several recent papers model human review of AI output. They include a Poisson escalation model with Erlang-C staffing (Besanson, 2026), an M/G/1 model for reviewing AI drafts (Bartolucci and Vivo, 2026) and a fluid-limit analysis of routing between an LLM judge and human reviewers (Lin and Zhang, 2026). All of them assume Poisson arrivals and rely on mean-based stability or waiting-time criteria. Taking a public-health approach to AI incident databases, Abraham et al. (2026) find strong over-dispersion in counts of vehicle and deepfake incidents, and Tran-Truong and Le (2026) show that their Markov reliability model of LLM agents reduces to the Goel-Okumoto software-reliability model when failures are rare. By contrast, we size capacity at a quantile, attach a price to it, and test the assumptions about arrivals.

2.5 Correlated risk in layered defences

In reliability engineering the multiplicative formula for the residual rate of independent safeguards is standard, and it is known to over-credit redundancy when failures are positively associated (Esary et al., 1967). The same issue arises in fault-tolerant software. Knight and Leveson (1986) tested and rejected the assumption that independently written program versions fail independently, and Eckhardt and Lee (1985) and Littlewood and Miller (1989) explain such coincident failures by variation in the difficulty of the inputs, so that diversity of design does not deliver independence of failure. This input-difficulty mechanism is the one we expect in guardrail stacks that face the same ambiguous outputs. For LLM defence stacks, Alotaibi et al. (2026) measure positive failure correlation across seven defence layers (φ between 0.30 and 0.75 on JailbreakBench) and describe the product as a lower bound that no real stack reaches. Bhardwaj et al. (2026) derive a Fréchet-Hoeffding sandwich and report high co-failure among components built on the same model. Our results add evidence from a different setting, hallucination-grounding layers, and we follow the correction through to the decision on capacity and cost. The literature on LLM guardrails moved fast in 2026, and since our search was not exhaustive, similar work may well exist.

3. Model

3.1 Environment and the firm's problem

Consider a firm that answers queries arriving at a time-varying rate $\Lambda(t)$ with a daily cycle. Each query produces a hallucination independently with probability p_0, and a cascade of screening layers may block some of these. We rely on three strong assumptions and return to them in Section 6.3. First, queries are independent conditional on the arrival intensity. Second, every hallucination that gets through the whole cascade is detected and turns into exactly one human-handled event, which takes up one unit of standby capacity. We refer to such an event as a critical hallucination and write $q \leq p_0$ for the probability that a query produces one. Third, one unit of capacity clears one critical event per day. The second assumption sets the information problem aside, since the firm sees every error that slips through. The model is therefore about the cost of absorbing errors, not the cost of finding them; Section 6.3 returns to this assumption.

The firm pays C_h per day for each unit of standby capacity and g_i per day for the compute needed to run screening layer i. It also incurs a false-block cost b for every correct answer that a layer wrongly rejects, and a shortfall cost C_u for every critical event that finds no capacity free on the day. The firm chooses capacity k and screening depth N to minimise expected daily cost. Following common practice, we first treat the service level, the probability α that capacity is sufficient, as a policy parameter; Section 3.6 then derives the shortfall cost that would justify it.

Under these assumptions, the thinning theorem (Kingman, 1993) implies that critical hallucinations follow a non-homogeneous Poisson process (NHPP) with intensity $q\Lambda(t)$. Let X_d denote the number of events on day d, whose mean over the day is $\lambda = q\int\Lambda(t)dt$. Then X_d is Poisson($\lambda$) regardless of how Λ is shaped within the day; that shape matters only for hourly staffing. If only a fraction δ of the surviving hallucinations is ever noticed and triaged, the event rate becomes $\delta\lambda$ and, by Proposition 3, the dispersion index becomes $1 + \delta\lambda c^2$. The capacities we report are therefore upper bounds, valid for $\delta = 1$. Under the NHPP, the time-rescaled gaps $\Lambda(t_i) - \Lambda(t_{i-1})$ are independent Exp(1) variables (Ogata, 1988; Daley and Vere-Jones, 2003), which gives the goodness-of-fit test we use in Section 5. If the daily rate is itself random, with $\Lambda_d = \lambda M$, $E[M] = 1$ and $\mathrm{Var}(M) = c^2$, then the count is mixed Poisson and $\mathrm{Var}(X_d) = \lambda\varphi$, where $\varphi = 1 + \lambda c^2$ is the dispersion index. This is the Cox, or doubly stochastic, alternative to the Poisson null (Cox, 1955; Cox and Isham, 1980).

Given a service-level target α, the capacity requirement is the generalised inverse of the count distribution:

$$k^*(\alpha, \lambda) = \min \{ k \in \mathbb{N}_0 : P(X_d \le k) \ge \alpha \} \quad (1)$$

Proposition 1 (Poisson capacity). Let X ~ Poisson(λ). (a) $k^*(\alpha, \lambda)$ is non-decreasing in λ. (b) $k^*(\alpha, \lambda) = 0$ if and only if $\lambda \le -\ln \alpha$; for α = 0.95 this threshold is 0.0513 events per day. (c) Provisioning at the mean, $k = \lfloor \lambda \rfloor$, leaves the system short with probability $P(X > \lfloor \lambda \rfloor)$, which is at least 0.26 for every $\lambda \ge 1$, about 0.35 at λ = 3, and tends to 0.5 as λ grows. (d) $k^*(\alpha, \lambda) = \lambda + z_\alpha \sqrt{\lambda} + O(1)$ as $\lambda \to \infty$.

Part (c) formalises certainty-equivalent staffing. A team sized at the mean load is short on between a quarter and a half of days, even when the Poisson assumption holds exactly. All proofs are given in Appendix A.

3.2 Demand uncertainty beyond Poisson

Proposition 2 (over-dispersion raises the safety stock). Let $X_d \mid M \sim \text{Poisson}(\lambda M)$ with E[M] = 1 and $\text{Var}(M) = c^2 > 0$. Then (a) $\text{Var}(X_d) = \lambda\varphi$ with $\varphi = 1 + \lambda c^2$. (b) X_d dominates Poisson(λ) in the convex order, so $E[(X_d - k)^+] \ge E[(Y - k)^+]$ for every k, where Y ~ Poisson(λ); in words, the expected shortfall at every capacity is higher than a Poisson model says. (c) If φ stays bounded as λ grows, $k^*_{\text{mixed}}(\alpha) = \lambda + z_\alpha \sqrt{\lambda\varphi} + O(1)$. If instead the multiplier has a fixed coefficient of variation c, then φ grows with λ and $k^*_{\text{mixed}}(\alpha) = \lambda \cdot F_M^{-1}(\alpha) + O(\sqrt{\lambda})$, where F_M is the distribution of M. In that regime the normal approximation understates the exact quantile by an amount that grows linearly in λ, and √φ is a lower guide for the inflation of the safety stock $k^* - \lambda$, not an exact factor.

Take a Gamma multiplier with λ = 120.7 and c = 0.3. The exact ratio of the safety stock to its Poisson value is then 3.7, compared with √φ = 3.4; at c = 0.44 the ratio is 5.5, compared with 4.9. Planning should therefore rely on the exact negative-binomial quantile (Table 1). With λ = 10 events per day and c = 0.3, the Poisson plan of $k^* = 15$ is exceeded on 11.0% of days rather than 4.9%, while the correct requirement is 18. With λ = 25 and c = 0.3, the Poisson plan of 33 is exceeded on 16.7% of days, and 41 units are needed.

Table 1. Capacity under over-dispersed arrivals (Gamma-mixed Poisson, α = 0.95)

λ (events/day)	c (CV of daily rate)	φ	k^* Poisson	k^* mixed	P(exceed) Poisson plan
10	0	1.00	15	15	4.9%
10	0.1	1.10	15	16	5.7%
10	0.2	1.40	15	17	8.0%
10	0.3	1.90	15	18	11.0%

10	0.5	3.50	15	21	16.4%
25	0	1.00	33	33	5.0%
25	0.1	1.25	33	35	7.0%
25	0.2	2.00	33	37	11.9%
25	0.3	3.25	33	41	16.7%
25	0.5	7.25	33	50	23.0%

Exact negative-binomial quantiles, computed by paper_tables.py.

3.3 Screening as thinning

Requests can arrive in clusters even when incidents are rare. Since critical hallucinations are an independent thinning of requests, the dispersion of the incident stream is inherited from the request stream.

Proposition 3 (thinning reduces dispersion). Let N be the request count in a window with mean μ and Fano factor F = Var(N)/μ, and let each request independently become a critical event with a constant probability p that does not depend on the request stream. Then the event count N_p has mean pμ and Fano factor 1 + p(F − 1). Equivalently, for a mixed-Poisson day with a rate multiplier of coefficient of variation c, $\varphi = 1 + \lambda c^2$ with λ = pμ, and independent thinning with constant p leaves M, and hence c, unchanged.

Screening thus has two effects at once. It reduces the mean number of events, and it pulls the stream closer to Poisson. Whereas φ changes under thinning, the coefficient of variation c does not, so c is the quantity that carries over between streams with different event rates. This holds provided p is constant and independent of M, which we assume without testing. In practical terms, demand uncertainty is most costly exactly where capacity is largest, that is, when screening is weak, and its cost falls once filtering is strong.

3.4 Correlated risk and the diversification fallacy

For a cascade of N layers with stand-alone clearance rates c_i (the share of critical hallucinations that layer i blocks), the residual rate is usually taken to be $\lambda_N = \lambda_0 \Pi(1 - c_i)$. This presupposes that the layers decide independently. In reality, the same ambiguous or adversarial outputs tend to fool every layer, so redundancy reduces risk by less than the formula suggests, much as in a portfolio of positively correlated assets.

Proposition 4 (dependence between layers). Let I_i indicate that layer i catches a given critical hallucination, with P(I_i = 1) = c_i, and let r = P(I_1 = … = I_N = 0), so that $\lambda_N = \lambda_0 r$. (a) (Sharp bounds) $(1 - \Sigma c_i)^+ \le r \le 1 - \max_i c_i$, and every value in this interval is attained by some joint distribution with the given marginals. If the I_i are positively associated (Esary et al.,

1967), then additionally r ≥ Π(1 − c_i). (b) (Two layers) r = (1 − c_1)(1 − c_2) + φ√(c_1(1 − c_1)c_2(1 − c_2)), where φ is the correlation of I_1 and I_2, so that r/[(1 − c_1)(1 − c_2)] = 1 + φ√(c_1c_2/[(1 − c_1)(1 − c_2)]). (c) (Common difficulty) Suppose the layers catch independently given a latent difficulty of the hallucination, with a common catch probability that is Beta-distributed with mean c, and let ρ be the correlation between any two catch indicators and s = (1 − ρ)/ρ. Then r_N = Π_{k=0}^{N−1} [(1 − c)s + k]/(s + k). This is non-decreasing in ρ for every N, equals (1 − c)^N as ρ → 0, tends to 1 − c as ρ → 1, and decays only polynomially, r_N ~ [Γ(s)/Γ((1 − c)s)] N^{−cs}, rather than geometrically. (d) Through Proposition 1(a), the required capacity k∗(α, λ_0r) is non-decreasing in r, and under (c) non-decreasing in ρ.

Part (a) says that the multiplicative formula is the special case of independence, not a bound: it understates residual risk, and hence required capacity, only when the layers are positively associated, which is what we expect when the same ambiguous outputs fool every layer. Part (b) shows how much a small correlation can matter when clearances are high. The multiplier √(c_1c_2/[(1 − c_1)(1 − c_2)]) grows with both clearances, so a correlation of 0.15 between layers with clearances of 0.895 and 0.58 raises the residual share by a factor of 1 + 0.15 × 3.43 = 1.51. Part (c) gives a closed form for a one-parameter family that runs from independence to total dependence. With c = 0.63 and ρ = 0.1, the residual rate after six layers is 5.4 times its value under independence, and extra layers keep paying off only slowly, because the decay is a power of N. The dependence must therefore be measured (Section 5.2) rather than assumed.

Table 2 shows how large the identification problem is. With c_i = 0.63 and λ_0 = 10, three layers could imply anything between 0 and 7 units of capacity, depending on a dependence that is unknown. Section 3.6 puts a price on this error in Table 7.

Table 2. Residual rate and capacity bounds (λ_0 = 10, c_i = 0.63, α = 0.95)

N	λ_N lower	λ_N independent	λ_N upper	k∗ lower	k∗ indep.	k∗ upper
1	3.70	3.70	3.70	7	7	7
2	0.00	1.37	3.70	0	4	7
3	0.00	0.51	3.70	0	2	7
4	0.00	0.19	3.70	0	1	7

3.5 Substituting compute for labour: the optimal depth of screening

Let k_min ≥ 0 denote a minimum standing capacity, reflecting the fact that an on-call engineer is not let go simply because incidents are rare. Under a service-level constraint, the firm's cost is

$f(N) = C_h \cdot \max\{ k_{min}, k^*(\alpha, \lambda_N) \} + \Sigma_{i \le N} g_i$ (2)

subject to a latency constraint $\Sigma_{i \le N} L_i \le SLA$, where L_i is the latency added by layer i. Screening acts as a capital-like input that reduces the amount of labour required, which makes this a problem of factor substitution.

Proposition 5 (optimal depth and factor substitution). Let $K(N) = \max\{k_{min}, k^*(\alpha, \lambda_N)\}$. (a) In general f is not unimodal in N: because k^* is integer-valued with plateaus, f can rise and then fall again. (b) (Breakpoints) Every minimiser is N = 0 or satisfies $K(N) < K(N-1)$, so it is enough to evaluate f at the first N of each capacity level. With $k_{min} \ge 1$, let N_0 be the first N with $k^*(\alpha, \lambda_N) \le k_{min}$; every $N > N_0$ is strictly dominated, so $N \le N_0$. (c) Let $\theta = C_h/g$ with equal per-layer compute cost g. The smallest and the largest minimiser are non-decreasing in θ: the dearer labour is relative to compute, the deeper the optimal cascade. (d) (Unknown dependence) If the layers have the same clearance c and nothing is known about their dependence, the worst-case cost of depth $N \ge 1$ is $C_hK(1) + gN$, attained when the layers fail together. It is strictly increasing in N, so the maximin depth is 0 or 1. The gain of depth N over depth 1 is $C_h[K(1) - K(N)] - g(N-1)$, and it is positive only if dependence is weak enough to save more than $g(N-1)/C_h$ units of capacity. (e) (Known dependence) Under the common-difficulty model of Proposition 4(c), $f(N; \rho)$ is non-decreasing in ρ for every N, so the minimal cost is non-decreasing in ρ, but the optimal depth need not be monotone in ρ (Table 5). (f) The latency constraint restricts N to a feasible prefix of the cascade.

Table 3 is based on an illustrative set of parameters: $\lambda_0 = 10$ events per day, a clearance of 0.63 per layer under independence, \$150 per capacity unit and \$25 per layer. With $k_{min} = 0$, cost falls between N = 3 and N = 4, rises at N = 5 and falls again at N = 6, the point at which k^* reaches zero ($\lambda_6 = 0.026 < -\ln 0.95$). The minimum is therefore at N = 6. If one on-call engineer must always be kept ($k_{min} = 1$), the minimum moves to N = 4. This non-monotone cost schedule is the signature of indivisibility. Because capacity comes in whole units, marginal analysis can only find local optima. Table 4 reports the comparative statics. As labour goes from costing the same as compute to costing many times more, the optimal depth increases from three layers to four or six and then levels off, because the floor and the integer plateau become binding. The cost model therefore does not support a fixed rule such as "three layers".

Table 3. Daily cost by guardrail depth ($\lambda_0 = 10$, c = 0.63 independent, C_h = \$150/unit, g = \$25/layer)

N	λ_N	k^*	Labour	Compute	Total (k_{min} = 0)	Total (k_{min} = 1)
0	10.000	15	\$2,250	\$0	\$2,250	\$2,250

1	3.700	7	$1,050	$25	$1,075	$1,075
2	1.369	4	$600	$50	$650	$650
3	0.507	2	$300	$75	$375	$375
4	0.187	1	$150	$100	$250	$250
5	0.069	1	$150	$125	$275	$275
6	0.026	0	$0	$150	$150	$300
7	0.009	0	$0	$175	$175	$325

Minimum: N = 6 under k_min = 0; N = 4 under k_min = 1. The cost parameters are illustrative assumptions.

Table 4. Optimal depth of screening against the labour-to-compute price ratio θ = C_h/g (λ_0 = 10, c_i = 0.63, α = 0.95, g = $25)

C_h ($/unit/day)	θ	N* (k_min = 0)	N* (k_min = 1)
25	1	3	3
50	2	4	4
100	4	6	4
150	6	6	4
300	12	6	4
600	24	6	4
1200	48	6	4

Computed by econ_tables.py; the cascade is evaluated under independence. Ties (θ = 1 and θ = 2 at k_min = 0) are reported at the smaller N.

Table 5 adds dependence to the example, using the common-difficulty model of Proposition 4(c). Dependence raises the cost of every depth, as part (e) of Proposition 5 says, and it moves the optimum in an irregular way: the cost-minimising depth with k_min = 0 is 6, 6, 7, 9, 6, 10, 6 layers for ρ = 0, 0.02, 0.05, 0.1, 0.2, 0.3 and 0.5, because the plateaus of k^* shift with ρ. A firm that plans for independence (six layers, or four when one engineer is always kept) therefore pays 33% more than necessary at ρ = 0.1, which is within the range of correlations we measure (Sections 5.2 and 5.7), and 45% more with the staffing floor. Under total dependence a single layer is optimal and the independence plan wastes five layers. In short, the benefit of stacking depends on a parameter that the clearance rates do not reveal.

Table 5. Optimal depth of screening under common-difficulty dependence (λ_0 = 10, c = 0.63 per layer, α = 0.95, C_h = $150, g = $25)

ρ	λ_3	k^* (N = 3)	N^* (k_min = 0)	Cost at N^*	N^* (k_min = 1)	Cost at N^*	Extra cost of independence plan (k_min = 0)	Extra cost of independence plan (k_min = 1)
0 (independent)	0.51	2	6	$150	4	$250	$0	$0
0.02	0.56	2	6	$150	4	$250	$0	$0
0.05	0.64	2	7	$175	4	$250	+$125 (+71%)	$0
0.1	0.78	2	9	$225	5	$275	+$75 (+33%)	+$125 (+45%)
0.2	1.06	3	6	$300	6	$300	$0	+$100 (+33%)
0.3	1.37	4	10	$400	10	$400	+$50	+$150

							(+12%)	(+38%)
0.5	2.00	5	6	$600	6	$600	$0	+$100 (+17%)
1 (comonotone)	3.70	7	1	$1,075	1	$1,075	+$125 (+12%)	+$75 (+7%)

Beta-distributed catch probability with mean 0.63 and correlation ρ between layers (Proposition 4(c)); ρ = 1 is the comonotone case. The independent-optimal depths are N = 6 (k_min = 0) and N = 4 (k_min = 1). The independence plan is the independent-optimal depth; its extra cost is f(N; ρ) at that depth minus the minimum over N. Computed by econ_dependence.py.

3.6 The implied cost of a shortfall, and the cost of misspecification

Up to this point (Sections 3.1 to 3.5) the service level α was taken as given. An economist would want to know what shortfall cost makes it optimal. Suppose each critical event that finds no capacity available costs the firm C_u, through delay, escalation or harm to the customer, say, and that each unit of capacity costs C_h whether it is used or not. The firm then minimises $\Phi(k) = C_h\, k + C_u\, E[(X - k)^+]$.

Proposition 6 (newsvendor service level). (a) Φ is convex in k, and its smallest minimiser is the α°-quantile of X with $\alpha^\circ = 1 - C_h/C_u$, so that a service-level target α is cost-minimising exactly when $C_u = C_h/(1 - \alpha)$. (b) If $C_u \le C_h$ the optimal capacity is zero. (c) If the firm plans with a Poisson model when the true count is mixed Poisson, its perceived cost curve lies below the true one at every k, by Proposition 2(b), and the plan it chooses is generally too small. (d) The minimal expected cost $\Phi^\circ(C_h)$ is concave in the capacity price, and $d\Phi^\circ/dC_h = k^\circ$ wherever the minimiser is unique: the shadow price of a unit of capacity is the capacity held.

With C_h = $150 per unit per day, a 95% service level is optimal if an unhandled event costs C_u = $3,000. A 90% level is optimal at $1,500, and a 99% level at $15,000. The common 95% target thus carries a specific and checkable claim about the cost of an unhandled event: twenty times the price of the unit that would have dealt with it. Section 4.4 uses a scenario value of $100 for the residual harm of each hallucination that reaches a user. Were that the shortfall cost, it would justify holding no standby capacity at all. The comparison makes clear that the two costs refer to different events, namely harm from an error the user receives and delay in handling an error the firm has already detected.

Table 6 puts a price on misspecification at C_u = $3,000. Suppose the true daily count has a rate whose coefficient of variation is 0.3. A firm that plans with a Poisson model then holds 15 units instead of 18 at λ = 10 and pays 9.4% more in expected cost than it needs to. At λ = 120.7 it holds 139 units instead of 189 and pays 43.6% more. Staffing at the mean does far worse, with

excess costs of roughly 102 to 121%. As Propositions 2 and 3 predict, the loss increases with both the rate and the dispersion.

Table 6. Expected daily cost of planning errors (C_h = $150, C_u = $3,000, optimal service level 95%)

λ (events/day)	c	k optimal (true)	k Poisson plan	k mean plan	Optimal cost	Excess: Poisson plan	Excess: mean plan
10	0.3	18	15	10	$3,070	+9.4%	+117%
10	0.44	20	15	10	$3,558	+20.7%	+121%
25	0.3	41	33	25	$6,983	+18.9%	+107%
25	0.44	47	33	25	$8,335	+36.2%	+116%
120.7	0.3	189	139	120	$31,659	+43.6%	+102%
120.7	0.44	221	139	120	$38,669	+65.3%	+114%

True counts are negative binomial (Gamma-mixed Poisson). Costs are evaluated under the true distribution. Computed by econ_tables.py.

Table 7 does the same for the diversification fallacy, using a cascade of N identical layers that each clear 63% of hallucinations (λ_0 = 10). If the firm plans as though the layers were independent when in fact they fail together (the extreme upper bound in Proposition 4), its shortfall cost is many times what it planned for. With three layers, the firm holds 2 units and expects to pay $351 a day. Under the comonotone upper bound the same plan costs $5,823 a day, whereas the optimum, with 7 units, costs $1,218. The upper bound is of course an extreme case, and the dependence we measure in Section 5.2 is much weaker. Even so, the exercise shows that deep cascades with strong dependence are where the independence assumption is most costly.

Table 7. Cost of assuming independent layers when they fail together (λ_0 = 10, clearance 0.63 per layer, C_h = $150, C_u = $3,000)

N	λ_N independent	λ_N upper bound	k planned (independence)	Cost if independence holds	Cost if layers fail together	Optimal k / cost if layers fail together
2	1.37	3.70	4	$650	$2,462	7 / $1,218
3	0.51	3.70	2	$351	$5,823	7 / $1,218
4	0.19	3.70	1	$200	$8,324	7 / $1,218

Poisson counts in both cases. The upper bound is the comonotone case of Proposition 4. Computed by econ_tables.py.

4. Empirical Evaluation: Design

4.1 Data, generators and ground truth

The data come from the validation set of SQuAD 2.0 (Rajpurkar et al., 2018), which combines answerable questions with questions that the accompanying passage cannot answer. We drew a single random sample of 300 questions (seed 0; 148 unanswerable) and gave the same sample to all three generators. The three runs are therefore paired on the same questions rather than being independent samples. Each model is given the passage and the question and

must reply either with a short answer or with the token UNANSWERABLE. A response counts as a hallucination if it is not an abstention and if either the question is unanswerable, or the question is answerable and the answer matches no gold answer (by normalised containment, or with a token-F1 of at least 0.6). Abstentions are never counted as hallucinations and skip the guardrails. The generators are Qwen2.5-0.5B-Instruct, SmolLM2-360M-Instruct and Qwen2.5-1.5B-Instruct (Qwen Team, 2024; Allal et al., 2025), all decoded greedily on a CPU. We chose small open models so that the whole pipeline could be run end to end with full control over how latency is measured. Their hallucination rates are higher than those of frontier models, and absolute rates should not be carried over (Section 6.3). A second, held-out sample of 1,000 questions is described in Section 5.7.

4.2 Guardrail layers

The three layers follow the familiar progression from deterministic rules to a trained classifier to a semantic judge, and each is implemented and run on every answer. Layer L1 uses regular expressions to catch e-mail addresses, national identifiers, and card and phone numbers. It also checks for leaked template tokens and degenerate repetition, and it blocks any answer that contains a number not found in the evidence passage. Layer L2 is an extractive question-answering verifier. A model trained on SQuAD 2.0 (deepset/roberta-base-squad2) answers the question from the passage, the score is the token-F1 between its span and the generator's answer (zero if it predicts no answer), and the output is blocked if the score falls below 0.5, a threshold we left untuned. Layer L3 is an LLM judge (Qwen2.5-1.5B-Instruct in every run of the first sample; Section 5.7 adds larger judges) that reads the passage, the question and the answer and replies SUPPORTED or UNSUPPORTED (Zheng et al., 2023). It is called only when an answer is routed to it, which, when L2 is present, means only when the L2 score is below 0.9.

Our first version of L2 was an NLI cross-encoder (DeBERTa-v3-small; He et al., 2021) that scored whether the passage entails "the answer to Q is A". It turned out to be uninformative, and after seeing the results we replaced it with the extractive verifier (Section 5.1). We did not use safeguard classifiers of the Llama Guard type, as these are designed for policy violations rather than factual grounding. All layers work from the same evidence passage, as in retrieval-grounded verification. Since every layer is run on every example, any subset or cascade can be evaluated offline from the recorded flags without assuming independence. For each call we log the flag, the score, the wall-clock latency and any exception.

Layers fail open. Any call that raises an exception or takes longer than 2 s counts as a system failure, which lets us estimate the failure rate rather than assume it. Its verdict is discarded, so the answer passes that layer, and the latency it adds is capped at 2 s. No call raised an exception, so every system failure was a judge call that took longer than 2 s; on our CPU set-up these made up 25 to 30% of judge calls. Section 5.4 reports a sensitivity analysis in which slow verdicts are waited for and credited.

4.3 Traffic replay and estimators

Because production logs of hallucinations are not available, we simulate arrivals while measuring outcomes. Queries arrive according to an NHPP with a 24-slot hour-of-day intensity, averaging 200 queries per day over 90 days. The default profile follows business hours, and a second profile uses the hour-of-day pattern measured in the BurstGPT production trace (Section 4.5). To represent bursty demand, a Cox variant multiplies each day's intensity by a Gamma factor with mean 1 and coefficient of variation c. We take c = 0.3 and c = 0.44 as scenario values. Once trend is removed and holiday and low-volume days are excluded, public ticket streams yield daily-rate coefficients of variation of 0.10 to 0.38 (0.25 to 0.41 with all days included; Section 5.6). The value c = 0.3 thus falls within that range and c = 0.44 lies above it. Real request arrivals from a public chatbot log (WildChat) give a net-of-trend coefficient of variation of 0.26 to 0.28 (Section 5.5). The value 0.44 is the smallest coefficient of variation found in any 14-day window of BurstGPT (Section 5.5), a trace that is far more variable overall. It is a floor for that trace, and it should not be compared directly with the net-of-trend ticket values.

Each arriving query replays an example drawn uniformly from the measured pool, and it becomes a critical event if it is a hallucination that survives the cascade (the detection assumption of Section 3.1). For each event stream we estimate $\hat{\lambda}$ as events per day and run four tests. These are an index-of-dispersion test on daily counts; a chi-square test of constant hourly intensity; a Kolmogorov-Smirnov test of raw inter-arrival times against an exponential, and of time-rescaled gaps under the fitted hourly intensity (here the Lilliefors-type p-value is optimistic); and a negative-binomial fit to daily counts. To capture uncertainty due to the finite evaluation pool, we bootstrap over examples with 300 draws and report 95% percentile intervals. Since all arrivals are generated from the same pool, the replay checks whether the counting

process has the claimed structure given the measured per-query outcomes. It cannot check whether real users' queries are independent, and this is the main limit on external validity.

4.4 Cascade selection and cost model

We evaluate L1, L2 and L3 and every combination of them, always applied in the order L1, L2, L3, as well as the case with no guardrail. The best cascade is the one that minimises $k^* + 0.02 \times$ (mean added latency in ms) + (system failures per day) among the cascades whose mean added latency stays within the stated budget (100 ms, or 500 ms where indicated). These weights are arbitrary and were not tuned. Their only purpose is to give a single ranking for Sections 5.3 and 5.4, and under the 500 ms budget it selects L1+L2 for all three generators.

Daily cost equals $C_h \cdot \max(k_{min}, k^*)$ plus the costs of compute, false blocks and residual harm. We set C_h at \$150 per capacity unit and compute at \$0.0001 per classifier call and \$0.002 per judge call. The false-block cost is \$1 per wrongly blocked answer, and the residual-harm cost is \$100 per hallucination that reaches a user. These figures are scenario values, not estimates. The *minimum-dollar cascade* of Section 5.4 minimises this cost. It may differ from the best cascade, since it ignores latency and system failures. A sensitivity grid varies daily traffic (50, 200 and 800 queries), the false-block cost (\$0, \$1, \$10) and the residual-harm cost (\$0, \$100, \$1,000), which gives 27 scenarios per generator. Sampling uncertainty is assessed with a pool bootstrap, which also yields paired intervals for cost differences between cascades.

4.5 Public production data

We test the counting-process assumptions on real systems using seven public sources. BurstGPT (Wang et al., 2025) contains 10.6 million requests to regional Azure OpenAI services over 231 days. WildChat-1M (Zhao et al., 2024) contains 838,000 timestamped conversations with ChatGPT models from a free research chatbot over 387 days, of which we use only the arrival times. A dataset of LLM service outages and incident reports (Chu et al., 2025) lists 457 incidents declared by three providers from March 2023 onwards. Three company ticket logs cover human triage: customer tickets from an international software company (Abdellatif, 2026), the incident log of a ServiceNow instance (Amaral et al., 2018) and the help-desk log of an Italian software company (Verenich, 2016). Lastly, a curated database of court decisions involving fabricated content (Charlotin, 2026) is the only dated log of hallucination incidents we were able to find, though it covers only the legal domain and suffers from strong detection and

reporting bias. Appendix B sets out the filters, the work-week and holiday rules, and the estimators. Two design choices deserve mention here. We report every ticket analysis twice, once for all work days and once without holidays and low-volume days, because isolated dips inflate a dispersion that a local mean cannot absorb. We also evaluate every Poisson plan in two ways: against a two-sided local mean, which is not a feasible forecast, and causally, with a trailing mean and a coefficient of variation estimated only from earlier days.

Every simulation and bootstrap draw is seeded and can be reproduced with the scripts that accompany the paper (Appendix D).

5. Findings

Sections 5.1 to 5.6 report the first sample of 300 questions and the public traces. Section 5.7 repeats the evaluation on 1,000 fresh questions with the same generators, adds two larger judges, and reports how far the first-sample results carry over.

5.1 Hallucination rates and guardrail layer performance

On the same 300 questions, Qwen2.5-0.5B hallucinates 60.3% of the time, SmolLM2-360M 65.0% and Qwen2.5-1.5B 47.0%, and between 75 and 90% of hallucinations are answers to unanswerable questions. Table 8 reports results by layer. The deterministic layer L1 catches only 0.7 to 5.1% of hallucinations. The task involves no personal data or template leaks and few fabricated numbers, so L1 does almost nothing, although it also adds almost no latency and few false blocks. The judge L3 blocks 48 to 59% of hallucinations and 17 to 24% of correct answers, and each call takes about 1.7 to 1.8 s on our CPU. Since 25 to 30% of judge calls run past the 2 s timeout and have their verdicts discarded, these clearance rates are lower than they would be if slow verdicts counted (Section 5.4). The judge is Qwen2.5-1.5B-Instruct in all runs, which means one arm is judged by its own generator; Section 5.7 tests whether this matters.

The NLI classifier we tried first produced a negative result. Its entailment score separated correct from hallucinated answers with an AUC of only 0.52 to 0.58, where 0.5 corresponds to chance. At the default threshold it blocked 87 to 88% of hallucinations, but also 85 to 90% of correct answers. The capacity savings it seemed to deliver were an artefact of blocking nearly everything. This is why we report the false-block rate alongside every capacity figure, since a reduction in capacity alone cannot tell a useful layer from one that blocks indiscriminately. We replaced the classifier with the extractive verifier (AUC 0.84 to 0.90) and re-ran every layer on

the stored answers. We should be clear that the switch was made after we had seen the results, that the verifier is evaluated on the same 300 questions, and that it is effectively a second solver of the task it checks, trained on SQuAD, which flatters it relative to a generic classifier. All figures below are based on it.

Table 8. Guardrail layers on the same 300 SQuAD 2.0 questions for each generator (CPU inference; latency re-measured without concurrent jobs)

Generator	p_0 (hallucination rate)	c_1 (L1)	c_2 (L2 QA)	c_3 (L3 judge)	L2 false-block (good answers)	L3 false-block	L2 mean ms	L3 mean ms (capped at 2 s)
Qwen2.5-0.5B	60.3%	2.8%	89.5%	58.0%	15.1%	24.4%	98	1759
SmolLM2-360M	65.0%	5.1%	96.9%	48.2%	70.9%	16.5%	97	1792
Qwen2.5-1.5B	47.0%	0.7%	81.6%	58.9%	15.4%	22.8%	95	1728

c_i is the stand-alone share of hallucinated answers the layer blocks. False-block is the share of correct, non-abstained answers blocked (denominators 119, 103 and 136). The L2 threshold (F1 < 0.5) was not tuned; SmolLM2 gives longer answers, which lowers its token-F1 agreement and inflates its L2 false-block rate.

5.2 Dependence between layers and the cost of routing

Among hallucinated answers, the catch decisions of L2 and L3 show a weak positive association ($\varphi = 0.15, 0.11, 0.12$ for the three generators; 95% bootstrap intervals [0.00, 0.29], [-0.02, 0.21], [-0.06, 0.28]; Fisher exact $p = 0.054, 0.213, 0.186$). This is the direction Proposition 4 predicts. Even so, the intervals include zero for two of the three generators, and the runs share the same questions, so the evidence is suggestive rather than conclusive. L1 catches only 1 to 10 hallucinations per generator, and its pairwise statistics ($\varphi \leq 0.09$) are uninformative.

Weak association is not the same as a small effect on residual risk. By Proposition 4(b), a correlation φ multiplies the independence prediction by $1 + \varphi\sqrt{(c_2c_3/[(1 - c_2)(1 - c_3)])}$, and the multiplier is large when clearances are high. With the clearances in Table 8 this gives 1.5, 1.6 and 1.3 for the three generators, which reproduces the ratios we observe (Figure 1, left); the match is exact because the identity is exact. In the L2+L3 cascade, the share of hallucinations that get through both layers is 1.6 to 7.6% according to the independence formula, computed from the marginals of the same sample, and 2.6 to 9.9% when both layers see every answer. These are 1.5, 1.6 and 1.3 times higher (with 12, 5 and 14 hallucinations passing; pool-bootstrap intervals [1.0, 2.0], [0.9, 2.1], [0.8, 1.7]), and the interval contains 1 for 2 of the three generators. The cost of dynamic routing, where only answers with low agreement are sent to the judge, is easier to see. Here the ratios increase to 2.1, 1.9 and 2.3 ([1.7, 2.6], [1.7, 2.2], [1.9, 3.0]), and the

number of answers that get through rises to 17, 6 and 25. Routing reduces judge calls by 89 to 92%. The price is that the judge no longer sees answers the verifier regards as consistent, and these include a sizeable share of the hallucinations the judge would otherwise have caught. Since the absolute numbers of passing answers are small, all of these estimates are imprecise. Section 5.7 repeats this analysis on 1,000 fresh questions.

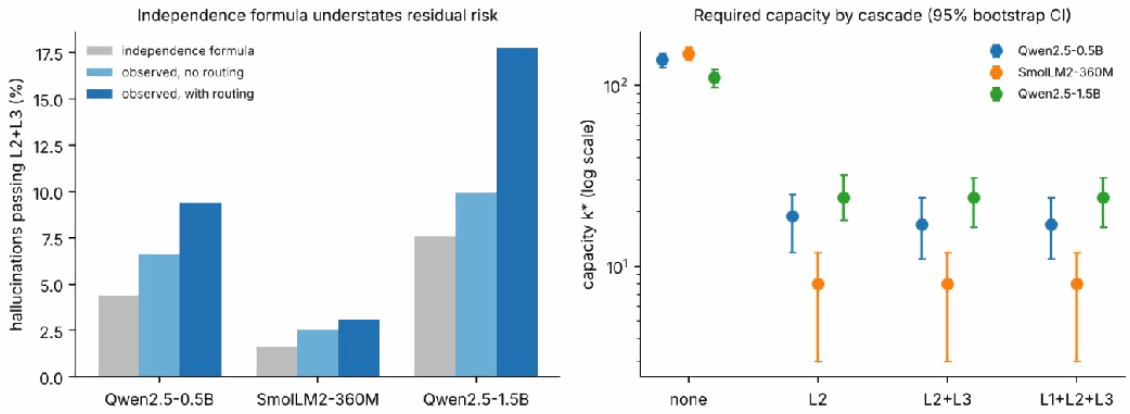


Figure 1. Left: share of hallucinations passing the L2+L3 cascade under independence, observed without routing, and observed with routing. Right: capacity k∗ by cascade (95% bootstrap CI over the evaluated pool), log scale.

5.3 Burstiness: Poisson-sized plans under Cox traffic

With non-homogeneous Poisson arrivals and no burstiness, the replay recovers the Poisson structure, as it must by construction. For the unprotected stream, the dispersion test on daily counts does not reject ($p = 0.53$ to 0.91). Raw inter-arrival times are clearly not exponential, because the intensity changes from hour to hour, but the time-rescaled gaps are consistent with Exp(1) ($p = 0.50$ to 0.85). This serves as a check on the replay and the estimators, and it confirms that the rescaling procedure behaves as the theory says it should. It says nothing about whether real traffic is Poisson, since the arrivals are simulated.

Once the daily intensity is multiplied by a Gamma factor whose coefficient of variation is just 0.3, the dispersion test rejects decisively ($p < 10^{-90}$ for the unprotected stream), and the negative-binomial shape parameter r falls between 12.7 and 13.4. Without guardrails, the Poisson capacities of $k* = 136$, 146 and 108 for the three generators are then exceeded on 23 to 28% of days, rather than the targeted 5%. The correct negative-binomial requirements are 178, 194 and 141, or 31 to 33% higher (Figure 2). Under the best cascade (L1+L2, with λ of roughly 4 to 17 per day) the gap in absolute capacity is smaller, but the Poisson plan is still exceeded on 10 to

11% of days (Table 9). Because we injected the burstiness ourselves, this replay confirms Proposition 2 and Table 1 on measured outcomes; it is not an independent empirical finding. Across every configuration we replayed, provisioning at the mean was exceeded on 28 to 49% of days, compared with the 5% target of the quantile plan.

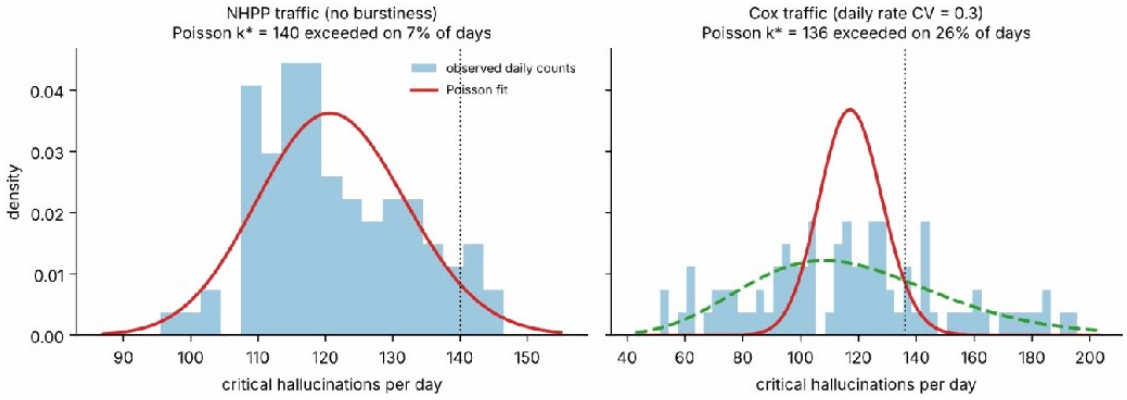


Figure 2. Daily critical hallucinations without guardrails (Qwen2.5-0.5B, 90 days, 200 queries/day). Left: steady NHPP traffic. Right: bursty traffic. Dotted line: Poisson k∗ (α = 0.95).

Some context helps in reading these scenario values. The smallest coefficient of variation in any 14-day window of BurstGPT is 0.44, which is a floor for a trace whose median window has 0.91 (Section 5.5). In public ticket streams, by contrast, the net-of-trend daily-rate coefficient of variation is 0.10 to 0.38 (Section 5.6; 0.25 to 0.41 before exclusions) and 0.26 to 0.28 in real chatbot request arrivals (Section 5.5). Table 9 reports exceedance rates under both scenario values, with the second combined with the BurstGPT hour-of-day profile. Guardrails narrow the gap without closing it. For the best cascade, the Poisson plan is exceeded on 7 to 16% of days at c = 0.44, against 30 to 32% without guardrails. This is what Propositions 2 and 3 would lead one to expect. For a given c, $\varphi = 1 + \lambda c^2$ is smaller when the event rate is lower, so over-dispersion adds fewer units to the safety stock (roughly 5, 1 and 8 extra), yet the probability of exceedance remains 1.3 to 3 times the target.

Table 9. Share of days on which the Poisson-sized capacity k∗ is exceeded (target 5%; 90 simulated days)

Generator	Cascade	Steady NHPP (c = 0)	c = 0.3	c = 0.44 + BurstGPT profile	k∗ Poisson → k∗ negative-binomial (c = 0.44)
Qwen2.5-0.5B	none	6.7%	25.6%	32.2%	135 → 209
Qwen2.5-0.5B	L1+L2	3.3%	10.0%	14.4%	18 → 23
SmolLM2-360M	none	5.6%	27.8%	32.2%	144 → 225
SmolLM2-360M	L1+L2	5.6%	11.1%	6.7%	8 → 9
Qwen2.5-1.5B	none	8.9%	23.3%	30.0%	106 → 162
Qwen2.5-1.5B	L1+L2	3.3%	10.0%	15.6%	24 → 32

Observed over 90 simulated days, so each share has a standard error of about 2 to 5 percentage points. L1+L2 is the best cascade by the objective of Section 4.4 within the 500 ms budget; it is not the minimum-dollar cascade of Section 5.4.

5.4 Capacity, latency and cost by cascade

For each generator, Table 10 reports the residual rate, the capacity, the added latency and the volume of false blocks for the main cascades at 200 queries per day, with bootstrap intervals over the evaluated pool. With a pool of 300 questions the intervals are wide, so differences of a few units of k^* between cascades are not significant, although the large differences are.

Table 10. Cascade performance at 200 queries/day, α = 0.95, SLA 500 ms mean added latency (95% bootstrap CI in brackets)

Generator	Cascade	λ per day	k^*	Mean added latency (ms)	False blocks / day	System failures / day
Qwen2.5-0.5B	none	120.7 [109, 132]	139 [126, 151]	0	0.0	0.0
Qwen2.5-0.5B	L1+L2	12.7 [7, 18]	19 [12, 25]	97	12.7	0.0
Qwen2.5-0.5B	L2	12.7 [7, 18]	19 [12, 25]	98	12.0	0.0
Qwen2.5-0.5B	L3	50.7 [39, 60]	63 [49, 73]	1759 ✗	19.3	60.0
Qwen2.5-0.5B	L2+L3	11.3 [7, 17]	17 [11, 24]	287	16.7	6.0
SmolLM2-360M	none	130.0 [120, 142]	149 [138, 162]	0	0.0	0.0
SmolLM2-360M	L1+L2	4.0 [1, 7]	8 [3, 12]	93	48.7	0.0
SmolLM2-360M	L2	4.0 [1, 7]	8 [3, 12]	97	48.7	0.0
SmolLM2-360M	L3	67.3 [58, 78]	81 [71, 94]	1780 ✗	11.3	50.0
SmolLM2-360M	L2+L3	4.0 [1, 7]	8 [3, 12]	249	50.7	6.7
Qwen2.5-1.5B	none	94.0 [82, 106]	110 [97, 123]	0	0.0	0.0
Qwen2.5-1.5B	L1+L2	17.3 [12, 24]	24 [18, 32]	87	14.0	0.0
Qwen2.5-1.5B	L2	17.3 [12, 24]	24 [18, 32]	88	14.0	0.0
Qwen2.5-1.5B	L3	38.7 [29, 48]	49 [39, 60]	1596 ✗	20.7	49.3
Qwen2.5-1.5B	L2+L3	16.7 [11, 23]	24 [16, 31]	279	19.3	6.7

L2+L3 denotes the routed cascade, in which the judge is called only when the L2 score is below 0.9, while L3 alone runs the judge on every answer. ✗ indicates that the 500 ms budget is exceeded. Mean added latency is computed per query and includes abstentions, which bypass the guardrails, so it differs from the per-layer means in Table 8. Each judge call is capped at the 2 s timeout, beyond which its verdict is discarded and the call counts as a system failure. Because L1 does almost nothing on this task, L1+L2 and L1+L2+L3 are equal to L2 and L2+L3 apart from small differences.

One task-specific verifier (L2) on its own brings required capacity down from 139, 149 and 110 units per day to 19, 8 and 24, a reduction of 78 to 95%. Adding the routed judge moves these figures to 17, 8 and 24, a difference of 2, 0 and 0 units. At the same time, it roughly triples the added latency and causes about six extra system failures per day through timeouts. Running the judge on every answer is not feasible under any of the latency budgets we consider (it costs about 1.6 to 1.8 s per query) and leads to 49 to 60 system failures per day. With a 100 ms budget, only L1 and L2 are feasible (L2 takes 90 to 100 ms here). With 500 ms, the routed judge fits on

average but not at the 95th percentile (1.8 to 1.9 s). On GPUs or hosted inference, latencies would be much lower.

Under the objective of Section 4.4, L1+L2 is the best cascade for all three generators. At the scenario costs, the minimum-dollar cascade is L1+L2+L3 for Qwen2.5-0.5B, L1+L2 for SmolLM2-360M and L1+L2+L3 for Qwen2.5-1.5B. Daily cost falls from $32,917, $35,350 and $25,900 without guardrails to $3,700, $1,649 and $5,286 at the minimum. For the three generators, the paired bootstrap interval for the daily saving from adding the routed judge to L1+L2 is $429 [-$5, $931]; -$2 [-$5, $0]; and $61 [-$9, $646]. The saving is positive but cannot be distinguished from zero for the two Qwen generators. For SmolLM2-360M it is slightly negative: the classifier leaves only 6 of its 195 hallucinations for the judge to catch, and the judge caught none of them. Under the fail-open policy, then, the judge does not demonstrably pay for itself for any of the generators.

This result hinges on the timeout policy. If slow verdicts were waited for and credited, L3's stand-alone clearance would be 63 to 84% rather than 48 to 59%, the routed cascade would need $k* = 15$, 8 and 21, and adding the judge would save $929 [$274, $1,861]; -$2 [-$5, $0]; and $709 [$130, $1,575]. Whatever value the judge appears to have therefore rests on accepting latencies longer than 2 s. A few further caveats apply. After L2, only a handful of hallucinations remain (19, 6 and 26 out of 300 questions); the 27-scenario grid varies cost parameters but does not reflect sampling error; and the runs share the same questions. None of the differences between generators should therefore be read as a general finding. What the data do support is that the classifier is the dominant layer, that the deterministic layer costs almost nothing but also achieves almost nothing on this task, and that the judge's marginal value is small and uncertain.

5.5 Evidence from public production traces

Request arrivals (BurstGPT). The Azure OpenAI trace contains 10.6 million requests over 231 recorded days, of which 4.6% failed (returned zero response tokens). The hour-of-day profile has a peak-to-trough ratio of 2.2, so arrivals are clearly non-homogeneous, in line with the NHPP formulation. Within an hour, counts vary more than Poisson would allow. The median variance-to-mean ratio of counts per bin (1 under Poisson) is 1.4 at 1 s, 5.3 at 10 s, 3.9 at 1 min and 5.1 at 10 min. Requests are therefore clustered at every scale from 1 s upwards, mildly at 1 s and strongly from 10 s on. By Proposition 3, an unprotected hallucination stream with $p = 0.6$

would inherit about 3.5 of this dispersion at the 10-minute scale, whereas a cascade with a residual probability of 0.05 would keep about 1.2.

Daily volume is extremely variable. The median day has 19,024 requests, but the range runs from 52 to 425,604. The median absolute change in log volume from one day to the next is 0.61, a factor of 1.8, and on 73% of days volume differs from the day before by more than 30%. Measured against the local level (the day's count divided by the mean of the 14 neighbouring days), the coefficient of variation is 1.22 over days 0 to 120 and 2.77 over days 225 to 334. Of the 163 14-day windows in which every day has more than 1,000 requests, the least variable has a coefficient of variation of 0.44, while the median window has 0.91. We therefore treat 0.44 as a floor on this trace's variability rather than an estimate comparable to the net-of-trend values in Section 5.6. The variability reflects traffic growth, campaigns, regime shifts and probably gaps in logging, as well as random fluctuation; the series also contains 7 near-duplicate pairs of days, which we treat as a data artefact. For this reason we use the trace as a stress test, not as an estimate of how dispersed hallucination counts are. In the stress test, we replay the measured multipliers, rescaled to have mean 1, on event streams averaging 10 and 120 events per day. With the multipliers from days 0 to 120, the Poisson plan is exceeded on 23% and 31% of days. With those from days 225 to 334, which regime shifts inflate, it is exceeded on 16% and 23%.

Real request arrivals (WildChat). The WildChat-1M corpus (Zhao et al., 2024) contains 837,989 conversations with ChatGPT models from a free research chatbot over 387 days, from 9 April 2023 to 29 April 2024, with a median of 2,053 per day. We use one timestamp per conversation, that of its last turn, as the arrival time of one request. The hour-of-day profile has a peak-to-trough ratio of 1.8 and weekends carry 0.91 times the weekday volume. Within an hour, counts are close to Poisson at short scales: on the 30 busiest days the median variance-to-mean ratio is 0.99 at 10 s and 1.01 at 1 min, and 1.29 at 10 min. Daily volume varies far less than in BurstGPT. The coefficient of variation of the daily rate net of trend and weekday is 0.28 (0.26 after excluding five low-volume days), and the least variable 14-day window has a coefficient of variation of 0.07 and the median window 0.14, against 0.44 and 0.91 in BurstGPT. This is of the same order as the ticket streams and close to our scenario value of $c = 0.3$. Replaying the measured multipliers on event streams with means of 10 and 120 per day, a Poisson-sized plan is exceeded on 7.0% and 19.1% of days in-sample and on 8.0% and 25.5% of days causally (net of trend: 7.5% and 19.2%, and 8.3% and 21.9%), against a target of 5%. These are request arrivals,

not hallucinations, from a single research chatbot. They show that real request streams lie within the range of variability we assumed, not what the variability of hallucination counts is.

Service incidents (LLM providers). From March 2023 to August 2024 the three providers declared 457 incidents, or 0.83 a day. The timing of incident starts is far from homogeneous. The average weekday count is 4.1 times the average weekend count, and homogeneity across hours of the day is rejected for every provider and subset. Every inter-arrival and dispersion test rejects a homogeneous Poisson model. Allowing for weekly and daily seasonality removes some of the misfit, but not all. The daily index of dispersion is 1.53, and it remains 1.30 after stratifying by day of week ($p = 6\times10^{-}6$); a negative-binomial fit to daily counts gives $r \approx 1.6$. Some of this over-dispersion is likely due to an incident rate that changed over the 18 months, which stratifying by weekday cannot remove. At such a low rate the practical consequence is small. The 95% capacities under Poisson and under the negative binomial are the same (3 incidents per day), and the Poisson capacity is exceeded on 3.5% of days, below the 5% target. This is consistent with Proposition 2, because $\varphi = 1 + \lambda c^2$ stays modest (about 1.2 to 1.5 here) when λ is below one per day.

We use incident duration, from the declared start to resolution, as a proxy for human response time. Its median is 1.0 h, its 90th percentile 5.1 h and its 99th percentile 34 h, with a mean of 3.1 h and a coefficient of variation of 3.5. Response times are thus heavy-tailed and far from exponential. The 95th percentile of the number of incidents open at the same time is 1, matching a Poisson occupancy with the same mean (1), and two or more incidents are open simultaneously 0.7% of the time. At provider level, then, one standing responder provides cover 95% of the time, and the rare long overruns come from the heavy tail of durations. These incidents are outages and degradations declared by providers, not hallucinations. They tell us about the arrival and service-time structure of an LLM incident stream, but not about its rate.

Dated hallucination cases (court decisions). The only public, dated log of hallucination incidents we could find is a curated database of court decisions in which a party relied on fabricated AI-generated content (Charlotin, 2026; 2,149 dated cases, 2023-04-14 to 2026-10-01). It covers only the legal domain, records the decision date to the day, and is subject to strong detection and reporting bias. Since counts rose from a handful a month to over a hundred in 2025, we analyse only the flatter stretch from October 2025 to August 2026 (1,527 cases, 6.3 per weekday) and leave out September and October 2026, as recent months are incomplete. Over

that period, weekday counts are over-dispersed relative to Poisson (day-of-week-stratified index 1.40, $p = 1\times10^{-}4$; net-of-trend daily-rate coefficient of variation 0.24). Part of this is due to the 10 weekdays that are US federal holidays. Without them the index drops to 1.25 and the coefficient of variation to 0.18, which is weak over-dispersion. At around six cases per weekday the effect on capacity is small. The 95% capacity is 11 under Poisson and 12 under a negative binomial with the stratified index, and a Poisson plan set against the local mean is exceeded on 6.3% of weekdays (5.6% with a causal plan). These results agree with Proposition 2 and with the other low-rate streams. Subject to the caveats above, this is to our knowledge the only dispersion estimate for dated hallucination incidents that can be obtained from public data.

5.6 Human triage workloads in public ticket data

The fourth family of public data is about what happens once an incident has been detected, that is, how tickets reach human teams in real organisations and how those teams work through them. These are customer-support and IT-incident tickets rather than LLM incidents. Still, they are the best public evidence we know of on arrival variability and on the distribution of service times in human triage. We analyse each stream for all work days and again with holiday and low-volume days removed under the rules in Appendix B; in Table 11, each cell reads "all days → excluded".

Table 11. Arrival structure of human-handled ticket streams (work days only; Poisson plan = 95% quantile of Poisson with the local mean)

Stream	Tickets (per work day)	Work-day-stratified dispersion index (includes slow trends)	Daily rate CV net of trend and weekday	Poisson plan exceeded: in-sample local mean	NB plan exceeded: in-sample	Poisson plan exceeded: causal	NB plan exceeded: causal
Software company (customer tickets 2017-23)	22,016 (13.1)	3.5 → 3.1	0.32 → 0.27	9.0% → 7.8%	3.4% → 4.0%	9.6% → 8.6%	1.9% → 2.5%
ServiceNow incidents (12-week segment, 2016)	22,035 (350.2)	25.2 → 5.9	0.25 → 0.10	35.0% → 20.0%	1.7% → 5.5%	15.0% → 6.7% (20 → 15 days)	0.0% → 0.0%
Italian software help desk (2010-12)	3,804 (5.3)	2.5 → 2.4	0.41 → 0.38	7.5% → 7.2%	3.7% → 3.8%	9.7% → 8.7%	2.9% → 2.8%

Each cell: all work days → low-volume or holiday days excluded (software company and ServiceNow: days below 35% of the rolling median, 62 and 5 days; Italian help desk: national holidays, 26 days). Target exceedance is 5%. The dispersion index includes slow trends, whereas the CV is net of a local level, so the two are not directly comparable. In-sample: the

local mean is a two-sided leave-one-out mean over about 40 surrounding work days after removing the weekday effect, and c is estimated from the same days that are scored, so neither is a feasible forecast. Causal: trailing 20-work-day mean, c estimated on earlier days only, rolling origin (the weekday effects are estimated on the full sample). The software company works Sunday to Thursday; the other work weeks are Monday to Friday.

Every stream is strongly non-homogeneous across the week and the day (weekend ticket rates are 0.09, 0.12 and 0.04 times the work-day rates). Arrivals stay far from Poisson even after time-rescaling, because daily volumes vary more than the weekly cycle explains. After removing slow trends and the weekday effect, the coefficient of variation of the daily rate is 0.32 → 0.27 for the software company, 0.41 → 0.38 for the Italian help desk and 0.25 → 0.10 for the ServiceNow segment, where holidays and bridge days account for most of the value. The dated hallucination cases of Section 5.5 give 0.24 → 0.18. These values are of the same order as our scenario values (c = 0.3 falls within the range after exclusions and c = 0.44 above it). We stress, however, that all of them come from non-LLM or legal-domain streams, and we make no claim to a universal value.

Since $\varphi = 1 + \lambda c^2$, the effect on capacity depends on the rate. Measured against the in-sample local mean, a Poisson-sized plan is exceeded on 7.8%, 7.2% and 20.0% of work days (at about 13, 5 and 350 tickets per day) instead of 5%, and on 9.0%, 7.5% and 35.0% when all days are included. A causal plan, the kind an operator could actually use, is exceeded on 8.6% and 8.7% of days for the two long streams with the Poisson quantile, and on 2.5% and 2.8% with a negative-binomial quantile. The negative-binomial plan is therefore somewhat conservative, for reasons these data do not allow us to identify. The in-sample negative-binomial columns serve as a calibration check on Proposition 2 rather than as an out-of-sample test, since c is estimated on the same days that are scored. The causal evaluation of the 12-week ServiceNow segment covers only 15 days and tells us nothing.

Service times are heavy-tailed (Appendix C, Table C1). Median resolution time is 168 h against a mean of 699 h for the software company, 22 h against 178 h for ServiceNow and 94 h against 211 h for the Italian help desk. The coefficients of variation are 3.0, 3.0 and 1.3, compared with 1 for an exponential distribution. Weibull shapes of 0.36 to 0.51 point to a decreasing hazard, meaning that a ticket that has already been open a long time is likely to stay open much longer. This has two implications for capacity planning. First, the data do not support models that assume exponential service. This is one more reason to size capacity from the quantile of the arrival count and to deal with the tail of service times explicitly, rather than through a mean-based utilisation. Second, consider the one stream with handling-time data, the

software company, where daily workload is the in-progress hours of the tickets created that day, attributed to the day of creation. There, the 95th percentile of daily workload is 3.2 times its mean, compared with 1.9 for the daily ticket count. Capacity measured in handling hours therefore needs a bigger multiple of the mean than capacity measured in arrivals, and the unit of triage capacity in Section 3.1 should be defined for a typical ticket, with the workload tail added on separately. Because in-progress time is calendar time spent in a workflow state, not effort, and because the comparison is not detrended, this ratio is only indicative.

For the software company and the Italian help desk, the time-averaged number of open tickets is 290 and 34. The observed 95th percentile of open tickets is 20% and 16% higher than for a Poisson occupancy with the same mean. This figure describes a static backlog plan sized on the long-run mean and does not test for Poisson arrivals, since volume in both logs shifts over time, and slowly varying volume alone is enough to produce such an excess.

5.7 Held-out replication

Two features of the first evaluation weaken it: L2 was chosen after the results had been seen, and the intervals are wide. We therefore repeated the evaluation on a fresh sample of 1,000 SQuAD 2.0 questions (a random draw with a fixed seed; overlap with the first 300 questions cannot be excluded and would be about 3% in expectation), with the same three generators and the same L1 and L2. The configuration was fixed and committed before any result was produced. The prompt of the judge L3 was not preserved from the first evaluation, so we rewrote it; L3 results are therefore not comparable with Table 8 and are reported separately. Three analysis details were fixed after seeing the outputs: the judge's reply was cut at four tokens, which turns UNSUPPORTED into UNSUPPORT for one model, and some models add newlines or explanations, so we classify a reply by its first SUPPORT or UNSUPPORT token; we added an exploratory variant in which slow verdicts are credited, as in Section 5.4; and, after seeing the results of the local judge, we added two larger judges from other families (below).

Table 12 shows the results. The verifier results replicate. L2 cuts required capacity by 84 to 96% (first sample 78 to 95%), and it again rejects many correct answers, 19 to 75% (first sample 15 to 71%). The false-block rate for Qwen2.5-0.5B is higher than in the first sample (24.0% against 15.1%, outside the first sample's interval), so the lower end of the earlier range should not be taken as typical. The dependence results replicate in direction and, with 1,000 questions, are no longer imprecise. The catch decisions of L2 and L3 are positively associated for all three

generators (φ = 0.18, 0.08 and 0.18), and the share of hallucinations that pass both layers is 1.24 to 1.33 times the independence prediction, with intervals that exclude one. With the local judge the magnitude is somewhat smaller than the 1.3 to 1.6 of the first sample, and with larger judges it is larger (below). By Proposition 4(b) this is what these correlations and clearances imply.

The judge is where the replication is most sensitive to the set-up. With the rewritten prompt and the local Qwen2.5-1.5B judge, L3 is much less active than in the first sample: it clears 19 to 25% of hallucinations and blocks 1 to 4% of correct answers, against 48 to 59% and 17 to 24% in Table 8; 26 to 31% of its calls exceed the 2 s timeout; and required capacity is the same with and without it (16, 6 and 18 units). Routing then changes nothing, because this judge flags almost no answer that L2 regards as consistent, so the routing effect of the first sample (a ratio of 1.9 to 2.3) is not reproduced by this judge. If slow verdicts are credited, L3 clears 27 to 34% of hallucinations, φ rises to 0.21, 0.10 and 0.23 and the ratio to 1.48, 1.39 and 1.38, but capacity is unchanged. In the cost model of Section 4.4, adding this judge changes the daily cost by less than one dollar, and slightly increases it.

To test whether the results depend on a judge that is its own generator, we ran two larger judges from other families through an API, Llama-3.3-70B-Instruct (Meta) and Gemma-3-27B-Instruct (Google), with the same prompt. Their latency over a network is not comparable with our CPU timings, so no verdict was discarded. Both are informative, and they change the picture. Gemma clears 54 to 69% of hallucinations and blocks 2 to 6% of correct answers, close to the profile of the judge in the first sample; Llama clears 78 to 86% and blocks 10 to 14%. With these judges the dependence is stronger than with the local judge and the routing effect reappears. The share of hallucinations that pass both layers is 1.6 to 2.0 times the independence prediction for Gemma and 2.5 to 3.3 times for Llama, and 2.1 to 2.5 and 3.8 to 5.0 times when answers with a high L2 score skip the judge, with all intervals above one; φ is 0.09 to 0.24 for Gemma and 0.14 to 0.31 for Llama. The ratio rises with the judge's clearance, which is what Proposition 4(b) implies: a given correlation is amplified when both layers catch most hallucinations. Adding the routed judge lowers required capacity by up to two units (for example from 16 to 14 for Qwen2.5-0.5B with Llama), and its daily saving in the cost model of Section 4.4, with a residual harm of $100, is $40 to $250 for Gemma and $209 to $669 for Llama. The intervals exclude zero in three of the six cases. A strong external judge therefore pays for itself modestly, whereas the

weak local judge does not, and the conclusion of the first sample that the routed judge does not demonstrably pay for itself holds for a weak or slow judge rather than in general.

A third local judge, SmolLM2-1.7B-Instruct, was uninformative. More than 90% of its calls exceeded the 2 s limit on our CPU, which makes it a no-op under the fail-open rule, and when its slow verdicts are credited it flags 99% of answers, correct and hallucinated alike. Its apparent capacity saving is, like that of the NLI classifier in Section 5.1, an artefact of blocking nearly everything.

Table 12. Held-out replication on 1,000 fresh SQuAD 2.0 questions (first-sample values in parentheses; 95% bootstrap intervals in brackets; local judge Qwen2.5-1.5B-Instruct with the rewritten prompt, slow verdicts discarded unless stated; the two large judges were run through an API and no verdict was discarded)

Measure	Qwen2.5-0.5B	SmolLM2-360M	Qwen2.5-1.5B
Hallucination rate, % (first sample)	61.9 (60.3)	65.8 (65.0)	47.8 (47.0)
L2 clearance, % (first sample)	91.3 (89.5)	97.7 (96.9)	87.4 (81.6)
L2 false-block, % (first sample)	24.0 (15.1)	75.1 (70.9)	18.7 (15.4)
k∗ without guardrails → with L2 (first sample)	142 → 16 (139 → 19)	151 → 6 (149 → 8)	112 → 18 (110 → 24)
Capacity reduction by L2, %	88.7	96.0	83.9
φ between L2 and L3 catch decisions (first sample 0.15, 0.11, 0.12)	0.18 [0.15, 0.21]	0.08 [0.06, 0.10]	0.18 [0.16, 0.21]
Passing both layers ÷ independence prediction (first sample 1.5, 1.6, 1.3)	1.33 [1.28, 1.39]	1.25 [1.21, 1.30]	1.24 [1.19, 1.29]
Same, slow judge verdicts credited (exploratory)	1.48 [1.39, 1.58]	1.39 [1.32, 1.46]	1.38 [1.31, 1.46]
Same, Gemma-3-27B judge (no timeout)	1.96 [1.64, 2.29]	1.91 [1.04, 2.69]	1.63 [1.40, 1.88]
Same, Llama-3.3-70B judge (no timeout)	2.66 [1.91, 3.56]	3.34 [1.49, 5.28]	2.53 [2.04, 3.16]
Routed cascade: Gemma-3-27B / Llama-3.3-70B	2.48 / 4.98	2.12 / 3.81	2.10 / 4.37
L3 clearance / false-block, %: Gemma-3-27B	61.2 / 3.2	68.5 / 5.8	54.0 / 2.1
L3 clearance / false-block, %: Llama-3.3-70B	84.0 / 10.0	86.0 / 13.7	78.2 / 9.8
k∗ with L2 → with L2+L3 (routed), local judge	16 → 16	6 → 6	18 → 18
L3 clearance / false-block, % (first sample)	24.7 / 2.9 (58.0 / 24.4)	20.1 / 1.3 (48.2 / 16.5)	19.2 / 3.8 (58.9 / 22.8)

Pre-registered configuration, code and per-question outputs are in the repository (Appendix D). Intervals resample questions (1,000 draws, seed 0).

6. Discussion

6.1 Economic interpretation

The model can be read as a standard problem of production under uncertainty, with the twist that the shocks come from the firm's own technology. The standby workforce is a safety

stock held against random errors. Its size is a quantile of the error count, pinned down by the ratio of the price of capacity to the cost of a shortfall (Proposition 6). Automated screening is a capital-like input that reduces the mean of what the workforce must absorb and, through thinning, its dispersion as well (Proposition 3). Labour and screening are thus substitutes, and the optimal mix depends on their relative price (Proposition 5). Framed this way, the operational question takes a price-theoretic form. When labour becomes more expensive, or inference cheaper, the firm should buy more screening, up to a limit set by the floor on standing staff and by the indivisibility of capacity.

Two features set this problem apart from textbook inventory theory. The first is that risk is correlated across screening layers. The independence formula gives redundancy credit for a diversification benefit that positive association takes away (Proposition 4), and the pricing exercise in Table 7 shows how expensive that mistake becomes when the layers fail together. The association we measured was modest. Even so, the share of hallucinations that passed two layers was higher than the independence prediction by a factor of 1.3 to 1.6 in the first sample, and in the held-out replication (Section 5.7) by 1.2 to 1.3 with the local judge and 1.6 to 3.3 with larger judges from other families, with intervals that exclude one. The second feature is that automation costs more than the compute bill. The verifier that cut required capacity by 78 to 96% also rejected 15 to 75% of correct answers. Part of the apparent saving is therefore a transfer of cost from the firm's standby labour to its customers and to its own revenue. A planner who reports capacity savings without the false-rejection rate overstates the benefit. The NLI classifier we tried first, which reduced capacity by blocking almost everything, is the extreme case.

The service level is itself an economic choice, not a technical constant. Proposition 6 shows that a 95% target is optimal only if an unhandled incident costs twenty times the price of a capacity unit, and that moving to 99% requires a shortfall cost five times higher ($15,000 rather than $3,000 at the scenario wage). Firms that quote service levels without stating a shortfall cost are therefore leaving the economics unspoken. The same proposition also puts a price on planning errors. In our examples, ignoring day-to-day variation in demand raised expected daily cost by 9.4% to 65.3%, and staffing at the mean load roughly doubled it. The second moment of demand is thus a first-order input to the staffing decision.

6.2 Implications for managers and regulators

For managers, the model suggests a practical sequence. Before fixing a staffing plan, estimate how dispersed incident counts are. In our replays a daily-rate coefficient of variation of 0.3 to 0.44 was enough for a Poisson-sized plan to be exceeded on 23 to 32% of days without guardrails and on 7 to 16% of days with the best cascade, against a 5% target. Make explicit the shortfall cost that the service-level target implies, and check that it is plausible. Judge each guardrail layer by its marginal effect on required capacity net of its false rejections, latency and failures, and measure how dependent the layers are before giving a stack credit for its nominal effectiveness. In the first sample, one verifier delivered most of the benefit, the deterministic layer did almost nothing, and the routed judge's saving could not be distinguished from zero (paired intervals for the daily saving $429 [-$5, $931]; -$2 [-$5, $0]; $61 [-$9, $646]) while roughly tripling latency. With larger external judges the saving was positive in some cases but not in others (Section 5.7). Auditing a sample of unflagged items is one way to measure what the screens are missing, because the count of handled incidents includes only the errors that were detected.

For regulators and auditors, the main lesson is about the evidence behind layered safeguards. Rules that require several independent safeguards should also require evidence that they are independent, because the product of stand-alone effectiveness is a lower bound on residual risk rather than an estimate of it. Liability rules feed directly into the model by setting the shortfall cost. A tenfold rise in the cost of an unhandled incident shifts the optimal service level from 90% to 99% (Proposition 6), and the standby capacity and screening depth that firms rationally choose shift with it. Lastly, if incident counts were reported publicly, with their dispersion and not just their totals, others could test the assumptions on which capacity planning depends. As we found, public data of this kind are almost non-existent.

6.3 Limitations and future research

The model's economic content also has limits. It solves a partial-equilibrium cost-minimisation problem for a single firm and is not a welfare analysis. The shortfall, false-block and residual-harm costs are scenario parameters, harms to users and wider external effects are folded into unit costs, and demand does not respond to quality or price. The model has no other strategic actors, so it leaves out the way adversaries adapt to known screens. Detection is

assumed to be complete, and reviewers' attention, which screening may wear down, is not modelled. Each of these is a natural extension, and each would alter the quantitative conclusions.

The most important limitation is that arrivals are simulated and that no operator log of hallucinations is publicly available. Outcomes are measured on real model outputs, but arrival times are generated and each arrival replays a bootstrapped example. The Poisson and time-rescaling checks in Section 5.3 therefore verify the machinery and hold partly by construction. The public traces are genuine production data, but what they record is requests, service incidents and human tickets, not hallucinations. The one dated hallucination log we found, a database of court decisions, is specific to one domain, resolves dates only to the day and is subject to strong detection and reporting bias. The burstiness in the replay is thus a scenario informed by real streams rather than a quantity estimated from them. Real request arrivals from a public chatbot log (Section 5.5) have a daily-rate variability in line with the scenario value of $c = 0.3$, but they are requests, not hallucinations. The most valuable next step would be validation on proprietary incident logs, and even a few months of daily counts of hallucination-related tickets would help. A closely related assumption is that every hallucination that passes the guardrails is detected and becomes one human-handled event. In practice only some are noticed or reported, so the capacities we report are upper bounds that hold under full detection (Section 3.1). The detection process deserves a model of its own.

The evaluation is small and in part post hoc. Because the three generators answered the same 300 questions, their results are paired rather than independent replications, and each layer estimate rests on 141 to 195 hallucinations and on anywhere from a few to a few dozen residual ones. The extractive verifier was brought in after the NLI classifier proved uninformative, so L2 was chosen after seeing results on the same data. It is trained on SQuAD, which flatters it, and its threshold was not tuned. The same 1.5B model serves as judge in every run of the first sample, so one arm judged itself. A held-out replication on 1,000 fresh questions with a configuration fixed in advance (Section 5.7) reproduces the verifier and dependence results, and two larger judges from other families show that the dependence does not come from a judge that is its own generator. It still uses the same task and models, the judge prompt had to be rewritten, and the judge-specific results (clearance, routing, saving) depend strongly on which judge is used. In the first sample the dependence estimates were imprecise: the association between L2 and L3 and the excess over the independence formula were weak and their bootstrap intervals included zero

or one for two of the three generators. In the replication the intervals exclude zero and one, but the independence prediction is still computed from the marginals of the same sample. The judge results also rest on the assumption that verdicts from calls slower than 2 s are discarded (Section 5.4). In addition, the evaluation relies on small open models of 0.4 to 1.5 billion parameters, a single extractive task and CPU latencies. Absolute hallucination rates are much higher than for frontier models, and latencies on GPUs or hosted inference would be different. In short, the method transfers but the numbers do not. Ground truth is based on string and token-F1 matching against SQuAD gold answers, which can flag valid paraphrases as hallucinations and so inflates the base rate. The cost parameters and selection weights are assumptions, so only the labour-to-compute ratio and the sensitivity grid carry information.

The public data come with caveats of their own. The ticket streams record customer-support and IT incidents, not LLM incidents. That hallucination tickets share their arrival variability and service times is an assumption. It is plausible for the coefficient of variation of the daily rate, which is unaffected by independent thinning with constant p, and less so for service times, which depend on the verification task. Resolution times are calendar times that include waiting on customers or vendors, so they do not measure triage effort. In two of the logs ticket volume shifts over time, the ServiceNow log supports only a 12-week segment, and the estimates of arrival variability rest on just three ticket streams. Isolated low-count days such as holidays and bridge days inflate a dispersion that a local mean cannot absorb. We report results both with and without them, using a volume rule for two streams and holiday calendars for two others, and we do not apply the outcome-based volume rule to low-rate streams. The causal negative-binomial plan turned out conservative for reasons we cannot explain, which tells us that our variability estimates are not fully understood. BurstGPT lacks calendar dates and contains a 104-day gap, near-duplicate pairs of days and order-of-magnitude regime changes in daily volume. Its day-level variability therefore reflects growth and logging as well as chance, and we use it only as a stress test.

We see three directions for future work. The first is validation on operators' production incident logs, which would make it possible to test the counting-process assumptions directly and to estimate, rather than assume, the dispersion of hallucination counts. The second is to model detection and triage explicitly, including reviewers' attention (which screening may wear down), the lag between an incident and its handling and heterogeneous failure classes with their own

arrival processes (clustered outages, heavy-tailed episodes), so that capacity can be expressed in terms of response-time targets as well as daily counts. The third is to examine how guardrail dependence varies with the generator, the domain and adversarial pressure, given that the evidence here comes from one task and three small models.

7. Conclusion

Human backup for machine errors is a safety stock, and screening is the capital that economises on it. Under the stated assumptions, several conclusions follow. The standby workforce should be set at a quantile of the daily incident count, with the service level implied by the cost of a shortfall. Over-dispersion raises the required safety stock above what a Poisson model or the dispersion index would suggest. The independence formula for stacked screens overstates diversification and understates residual risk, and because high clearances amplify even weak correlation, depth beyond one layer is justified only by an assumption about dependence. And when capacity is indivisible and standing staff has a floor, the cost-minimising depth of screening is not fixed at three but responds to the relative prices of labour and compute.

On the measured outputs of three small open models, a single task-specific verifier cut required capacity by 78 to 96% while rejecting 15 to 75% of correct answers, and a routed judge did not demonstrably pay for itself under a 2 s fail-open timeout with a weak local judge, although larger judges from other families did so modestly. In a held-out replication on 1,000 fresh questions the verifier results held, and the share of hallucinations passing two layers exceeded the independence prediction by 1.2 to 3.3 times, more as the clearance of the layers rose. Mild burstiness raised the exceedance rate of a Poisson-sized plan from 5% to about 23 to 32% without guardrails and 7 to 16% with the best cascade, and ignoring it raised expected cost by 9 to 65% in our examples. Public production data support the assumed structure of the arrival processes: requests arrive in clusters, incidents are seasonal with a drifting level, ticket streams have a net-of-trend daily-rate coefficient of variation of 0.10 to 0.38, and a public chatbot log has 0.26 to 0.28. Hallucination arrivals themselves are simulated, however, and no operator log is public. The key next step is therefore to test these assumptions on production incident logs and to estimate the shortfall cost directly.

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Data availability: The public datasets analysed (BurstGPT, the LLM outage and incident dataset, the three ticket logs, the WildChat-1M corpus and the AI Hallucination Cases database) are available from their original sources under the licences stated in Section 4.5 and Appendix B. Per-query model outputs and guardrail decisions are available at [repository URL].

Code availability: Analysis code is available at [repository URL].

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References

Abdellatif M (2026) Help Desk Tickets (Version 3). Mendeley Data (https://doi.org/10.17632/btm76zndnt.3).

Abraham S, Chen T, Chhun C, Jaramillo-Gutierrez G, Mylius S, Raaj S, Slattery P, McGregor S (2026) AI Incident Monitoring Through a Public Health Lens. arXiv:2604.19914.

Aksin Z, Armony M, Mehrotra V (2007) The Modern Call Center: A Multi-Disciplinary Perspective on Operations Management Research. Production and Operations Management 16(6):665–688.

Allal LB, Lozhkov A, Bakouch E, Blázquez GM, Penedo G, Tunstall L, Marafioti A, Kydlíček H, Piqueres Lajarín A, Srivastav V, Lochner J, Fahlgren C, Nguyen XS, Fourrier C, Burtenshaw B, Larcher H, Zhao H, Zakka C, Morlon M, Raffel C, von Werra L, Wolf T (2025) SmolLM2: When Smol Goes Big—Data-Centric Training of a Small Language Model. arXiv:2502.02737.

Alotaibi A, Jabbar MS, Al-Azani S, Ahmed M (2026) Layered LLM Defenses as an Ensemble: Access Tiers, Inference Cost, and the Measured Failure Correlation Between Defense Layers. arXiv:2608.28327.

Amaral CAL, Fantinato M, Peres SM (2018) Incident Management Process Enriched Event Log. UCI Machine Learning Repository (dataset 498).

Arrow KJ, Harris T, Marschak J (1951) Optimal Inventory Policy. Econometrica 19(3):250–272.

Avramidis AN, Deslauriers A, L'Ecuyer P (2004) Modeling Daily Arrivals to a Telephone Call Center. Management Science 50(7):896–908.

Bartolucci S, Vivo P (2026) Queue & AI: When Faster Tasks Slow Down the Workflow. arXiv:2605.27202.

Bassamboo A, Randhawa R, Zeevi A (2010) Capacity Sizing Under Parameter Uncertainty: Safety Staffing Principles Revisited. Management Science 56(10):1668–1686.

Berente N, Gu B, Recker J, Santhanam R (2021) Managing Artificial Intelligence. MIS Quarterly 45(3):1433–1450.

Besanson G (2026) SARC: A Governance-by-Architecture Framework for Agentic AI Systems. arXiv:2605.07728.

Bhardwaj VP, Singh G, Bhardwaj AP (2026) Agent Behavioral Contracts II: Certifying Compositional Reliability Without Assuming Independence. arXiv:2608.12895.

Borst S, Mandelbaum A, Reiman M (2004) Dimensioning Large Call Centers. Operations Research 52(1):17–34.

Brown L, Gans N, Mandelbaum A, Sakov A, Shen H, Zeltyn S, Zhao L (2005) Statistical Analysis of a Telephone Call Center: A Queueing-Science Perspective. Journal of the American Statistical Association 100(469):36–50.

Cameron AC, Trivedi PK (1998) Regression Analysis of Count Data. Cambridge University Press, Cambridge.

Charlotin D (2026) AI Hallucination Cases Database. (https://www.damiencharlotin.com/hallucinations/; CC BY 4.0; accessed October 6, 2026).

Chu X, Talluri S, Lu Q, Iosup A (2025) An Empirical Characterization of Outages and Incidents in Public Services for Large Language Models. In: Proceedings of ICPE 2025 (arXiv:2501.12469).

Cox DR (1955) Some Statistical Methods Connected with Series of Events. Journal of the Royal Statistical Society, Series B 17(2):129–164.

Cox DR, Isham V (1980) Point Processes. Chapman & Hall, London.

Daley DJ, Vere-Jones D (2003) An Introduction to the Theory of Point Processes, Volume I (2nd ed.). Springer, New York.

Eckhardt D, Lee L (1985) A Theoretical Basis for the Analysis of Multiversion Software Subject to Coincident Errors. IEEE Transactions on Software Engineering 11(12):1511–1517.

Esary JD, Proschan F, Walkup DW (1967) Association of Random Variables, with Applications. Annals of Mathematical Statistics 38(5):1466–1474.

Gallego G, Moon I (1993) The Distribution Free Newsboy Problem: Review and Extensions. Journal of the Operational Research Society 44(8):825–834.

Gans N, Koole G, Mandelbaum A (2003) Telephone Call Centers: Tutorial, Review, and Research Prospects. Manufacturing & Service Operations Management 5(2):79–141.

Green LV, Kolesar PJ, Whitt W (2007) Coping with Time-Varying Demand When Setting Staffing Requirements for a Service System. Production and Operations Management 16(1):13–39.

Halfin S, Whitt W (1981) Heavy-Traffic Limits for Queues with Many Exponential Servers. Operations Research 29(3):567–588.

Harrison JM, Zeevi A (2005) A Method for Staffing Large Call Centers Based on Stochastic Fluid Models. Manufacturing & Service Operations Management 7(1):20–36.

He P, Gao J, Chen W (2021) DeBERTaV3: Improving DeBERTa Using ELECTRA-Style Pre-Training with Gradient-Disentangled Embedding Sharing. arXiv:2111.09543.

Inan H, Upasani K, Chi J, Rungta R, Iyer K, Mao Y, Tontchev M, Hu Q, Fuller B, Testuggine D, Khabsa M (2023) Llama Guard: LLM-Based Input-Output Safeguard for Human-AI Conversations. arXiv:2312.06674.

Ji Z, Lee N, Frieske R, Yu T, Su D, Xu Y, Ishii E, Bang YJ, Chen D, Dai W, Chan HS, Madotto A, Fung P (2023) Survey of Hallucination in Natural Language Generation. ACM Computing Surveys 55(12): Article 248.

Jongbloed G, Koole G (2001) Managing Uncertainty in Call Centres Using Poisson Mixtures. Applied Stochastic Models in Business and Industry 17(4):307–318.

Kingman JFC (1993) Poisson Processes. Oxford University Press, Oxford.

Knight JC, Leveson NG (1986) An Experimental Evaluation of the Assumption of Independence in Multiversion Programming. IEEE Transactions on Software Engineering 12(1):96–109.

Lewis P, Perez E, Piktus A, Petroni F, Karpukhin V, Goyal N, Küttler H, Lewis M, Yih W, Rocktäschel T, Riedel S, Kiela D (2020) Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. Advances in Neural Information Processing Systems 33.

Lin R, Zhang J (2026) When to Screen, When to Bypass: LLM-Judges in Resource-Scarce AI-Human Workflow. arXiv:2603.13870.

Littlewood B, Miller D (1989) Conceptual Modeling of Coincident Failures in Multiversion Software. IEEE Transactions on Software Engineering 15(12):1596–1614.

Manakul P, Liusie A, Gales MJF (2023) SelfCheckGPT: Zero-Resource Black-Box Hallucination Detection for Generative Large Language Models. In: Proceedings of EMNLP 2023.

Ogata Y (1988) Statistical Models for Earthquake Occurrences and Residual Analysis for Point Processes. Journal of the American Statistical Association 83(401):9–27.

Porteus EL (2002) Foundations of Stochastic Inventory Theory. Stanford University Press, Stanford.

Qwen Team (2024) Qwen2.5 Technical Report. arXiv:2412.15115.

Raisch S, Krakowski S (2021) Artificial Intelligence and Management: The Automation-Augmentation Paradox. Academy of Management Review 46(1):192–210.

Rajpurkar P, Jia R, Liang P (2018) Know What You Don't Know: Unanswerable Questions for SQuAD. In: Proceedings of ACL 2018, pp 784–789.

Steckley SG, Henderson SG, Mehrotra V (2009) Forecast Errors in Service Systems. Probability in the Engineering and Informational Sciences 23(2):305–332.

Topkis DM (1978) Minimizing a Submodular Function on a Lattice. Operations Research 26(2):305–321.

Tran-Truong PT, Le XB (2026) Measuring the Unmeasurable: Markov Chain Reliability for LLM Agents. arXiv:2604.24579.

Van Mieghem JA (2003) Capacity Management, Investment, and Hedging: Review and Recent Developments. Manufacturing & Service Operations Management 5(4):269–302.

Verenich I (2016) Helpdesk. (help desk log of an Italian software company), Mendeley Data, V1 (https://doi.org/10.17632/39bp3vv62t.1).

Wang Y, Chen Y, Li Z, Kang X, Fang Y, Zhou Y, Zheng Y, Tang Z, He X, Guo R, Wang X, Wang Q, Zhou AC, Chu X (2025) BurstGPT: A Real-World Workload Dataset to Optimize LLM Serving Systems. In: Proceedings of KDD 2025 (arXiv:2401.17644).

Whitt W (2006) Staffing a Call Center with Uncertain Arrival Rate and Absenteeism. Production and Operations Management 15(1):88–102.

Whitt W (2007) What You Should Know About Queueing Models to Set Staffing Requirements in Service Systems. Naval Research Logistics 54(5):476–484.

Zhao W, Ren X, Hessel J, Cardie C, Choi Y, Deng Y (2024) WildChat: 1M ChatGPT Interaction Logs in the Wild. In: Proceedings of ICLR 2024 (arXiv:2405.01470).

Zheng L, Chiang WL, Sheng Y, Zhuang S, Wu Z, Zhuang Y, Lin Z, Li Z, Li D, Xing EP, Zhang H, Gonzalez JE, Stoica I (2023) Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. Advances in Neural Information Processing Systems 36.

Appendix A. Proofs

Proof of Proposition 1. (a) The Poisson(λ) distribution is stochastically increasing in λ, so P(X ≤ k) is non-increasing in λ and the smallest k that reaches α cannot fall as λ rises. (b) $k* = 0$ exactly when P(X = 0) = e^{−λ} ≥ α, that is, when λ ≤ −ln α. (c) For λ in [n, n + 1) with n ≥ 1, P(X > n) increases in λ, so its minimum on that interval is P(Poisson(n) > n). The sequence P(Poisson(n) > n) increases in n towards 1/2, and its first term is P(Poisson(1) > 1) = $1 - 2e^{-1} = 0.264$. At λ = 3 it equals 1 − P(X ≤ 3) = 0.353. (d) is the Cornish-Fisher expansion of the Poisson quantile. ∎

Proof of Proposition 2. (a) is the law of total variance: $\mathrm{Var}(X_d) = E[\lambda M] + \mathrm{Var}(\lambda M) = \lambda + \lambda^2 c^2$. (b) For Poisson Y_λ, $\partial E[(Y_\lambda - k)^+]/\partial\lambda = P(Y_\lambda \geq k)$, which is non-decreasing in λ, so $\lambda \mapsto E[(Y_\lambda - k)^+]$ is convex. Jensen's inequality applied to the mixing variable gives $E[(X_d - k)^+] = E[g(\lambda M)] \geq g(\lambda)$, where $g(\lambda) = E[(Y_\lambda - k)^+]$. (c) If φ is bounded, the count is asymptotically normal with variance λφ. For fixed c, X_d/λ converges in distribution to M as λ grows, so the α-quantile of X_d is asymptotically $\lambda F_M^{-1}(\alpha)$, and the normal approximation fails by an amount linear in λ. ∎

Proof of Proposition 3. Under binomial thinning with constant probability p, $\mathrm{Var}(N_p) = p^2\mathrm{Var}(N) + p(1 - p)\mu$ by the law of total variance. Dividing by $E[N_p] = p\mu$ gives $pF + (1 - p) = 1 + p(F - 1)$. The statement about c follows because thinning multiplies the intensity of each day

by p and leaves the multiplier M unchanged. We also verified the identity by simulation (checks/check_thinning.py). ∎

Proof of Proposition 4. (a) The bounds are the Fréchet-Bonferroni bounds for the probability that no layer fires. They are attained: nested events {I_i = 1} give r = 1 − max_i c_i, and events that are as disjoint as possible, covering the whole space when Σc_i ≥ 1, give r = (1 − Σc_i)⁺. The joint laws with fixed marginals form a convex set and r is linear in the joint law, so every intermediate value is attained. If the indicators are positively associated, P(∩{I_i = 0}) ≥ ΠP(I_i = 0) (Esary et al., 1967). (b) With $p_{11} = P(I_1 = I_2 = 1) = c_1c_2 + \mathrm{Cov}(I_1, I_2)$, $r = 1 - c_1 - c_2 + p_{11} = (1 - c_1)(1 - c_2) + \mathrm{Cov}(I_1, I_2)$, and $\mathrm{Cov}(I_1, I_2) = \varphi\sqrt{(c_1(1 - c_1)c_2(1 - c_2))}$. (c) Given the difficulty, the layers fail to catch independently, so r_N = E[(1 − π)^N] with π ~ Beta(a, b), a = cs, b = (1 − c)s. Then E[(1 − π)^N] = B(a, b + N)/B(a, b) = Π_{k=0}^{N−1} (b + k)/(a + b + k), and Corr(I_i, I_j) = Var(π)/(c(1 − c)) = 1/(a + b + 1), which equals ρ when a + b = s. The factor for k is 1 − cs/(s + k) = 1 − c/(1 + k/s); it equals 1 − c for k = 0 and is decreasing in s for k ≥ 1, hence non-decreasing in ρ, and so is the product. The limits follow from s → ∞ and s → 0, and the asymptotics from Γ(b + N)/Γ(a + b + N) ~ N^{−a}. (d) follows from Proposition 1(a). ∎

Proof of Proposition 5. (a) Table 3 is a counterexample to unimodality. (b) If K(N) = K(N − 1) then f(N) = f(N − 1) + g > f(N − 1), so N is not a minimiser; hence a minimiser is 0 or the first N of its capacity level. For $N > N_0$ the labour cost is constant at C_hk_min while compute cost increases in N, so such N are dominated by N_0. (c) Write f(N) = g[N + θK(N)] with K(N) non-increasing in N. The difference f(N + 1) − f(N) = g[1 + θ(K(N + 1) − K(N))] is non-increasing in θ, so f has decreasing differences in (N, θ) and both the smallest and the largest minimiser are non-decreasing in θ by Topkis' (1978) monotone comparative statics. (d) By Proposition 4(a), $\lambda_N \le \lambda_0(1 - c) = \lambda_1$ for every dependence structure, with equality when the layers fail together, so K(N) ≤ K(1) with equality in the worst case by Proposition 1(a). The worst-case cost of depth N ≥ 1 is therefore C_hK(1) + gN, which is strictly increasing in N, and the gain over depth 1 is f(1) − f(N) = C_h[K(1) − K(N)] − g(N − 1). (e) By Proposition 4(c), λ_N(ρ) is non-decreasing in ρ, hence so are K(N; ρ) and f(N; ρ); a minimum of non-decreasing functions is non-decreasing. Table 5 shows that the minimiser is not monotone. (f) is immediate. ∎

Proof of Proposition 6. (a) For integer k, E[(X − k − 1)⁺] − E[(X − k)⁺] = −P(X > k), so Φ(k + 1) − Φ(k) = C_h − C_uP(X > k). The right-hand side is non-decreasing in k, so Φ is convex, and the smallest minimiser is the smallest k with P(X > k) ≤ C_h/C_u, that is, P(X ≤ k) ≥ 1 −

C_h/C_u. Setting $\alpha = 1 - C_h/C_u$ gives $C_u = C_h/(1-\alpha)$. (b) If $C_u \le C_h$ then $\Phi(1) - \Phi(0) = C_h - C_u P(X > 0) \ge C_h - C_u \ge 0$, and convexity gives $k^\circ = 0$. (c) By Proposition 2(b) the stop-loss transform of the mixed-Poisson count is at least that of the Poisson count at every k, so a firm that uses the Poisson model underestimates the shortfall term of Φ at every capacity and, because the marginal benefit of capacity $C_u P(X > k)$ is perceived to be lower where the tail is lighter, tends to choose a smaller k; Table 6 reports the resulting cost. (d) $\Phi^\circ(C_h) = \min_k [C_h k + C_u E(X - k)^+]$ is a minimum of functions linear in C_h, hence concave, and by the envelope theorem its derivative at a point where the minimiser is unique equals that minimiser. ∎

Appendix B. Public Data: Sources, Filters and Estimators

BurstGPT. The trace (Wang et al., 2025; CC BY 4.0) holds 10.6 million request records of regional Azure OpenAI ChatGPT and GPT-4 services at one-second resolution over 231 days, in two contiguous segments of 121 and 110 days separated by an unrecorded gap of 104 days. Timestamps are seconds from an undisclosed day 0, so weekday labels are unknown. Failed requests are rows with zero response tokens. We estimate the hour-of-day profile, the day-to-day variability of request volume (robustly, because volume grows by two orders of magnitude and contains regime shifts) and the within-hour Fano factor of counts at 1 s, 10 s, 1 min and 10 min. The stress test replays the measured daily multipliers, rescaled to mean 1, on event streams with a given mean.

Provider incidents. The dataset of LLM service outages and incident reports (Chu et al., 2025; CC BY 4.0) contains provider-declared incidents scraped from the status pages of OpenAI, Anthropic and Character.AI. We use the 457 incidents starting in March 2023, when all three pages are populated. We test Poisson counts per day and week (index of dispersion, also stratified by day of week), day-of-week and hour-of-day homogeneity, exponential inter-arrival times after time-rescaling with a fitted hour-of-week intensity, the distribution of incident durations as a proxy for human response time, and the number of concurrently open incidents against the M/G/∞ prediction.

Ticket logs. The software-company data (Abdellatif, 2026) contain 66,691 rows; we keep tickets of type "Ticket" created in 2017 or later and resolved as "Done", which leaves 22,016 tickets with creation, resolution and in-progress times. The work week is detected from the data as the five busiest days (Sunday to Thursday for this company, Monday to Friday for the others).

The ServiceNow log (Amaral et al., 2018; 24,918 incidents in 2016 to 2017) has 98.9% of its incidents opened between 29 February and 1 June 2016, so its arrival analysis uses only the middle weeks (7 March to 27 May 2016) and its concurrency is not analysed. The Italian help-desk log (Verenich, 2016) contains 3,804 cases from 2010 to 2012. A High/Highest/Blocker subset of the software company is not analysed because its rate rises from about zero to eleven tickets per work day over 2017 to 2023, which reflects a change in the priority field.

Holiday and low-volume exclusions. Holidays and bridge days produce isolated low-count days that a local mean cannot absorb, so every arrival analysis is reported twice. For the software company and ServiceNow we exclude work days with fewer than 35% of the centred 41-work-day rolling median (62 and 5 days), a rule that is reliable only for high-rate streams. For the Italian help desk we exclude Italian national holidays and Easter Monday (26 days), and for the court cases US federal holidays (10 days).

Estimators. For each stream we estimate weekday-stratified over-dispersion, the daily-rate coefficient of variation net of slow trends (the Pearson dispersion of weekday counts against a leave-one-out local mean over about eight weeks, after removing the weekday effect), the exceedance rate of Poisson-sized capacity plans, the distribution of resolution and in-progress times (exponential, lognormal and Weibull fits compared by AIC), the daily workload safety factor (95th percentile over mean of handling hours, against arrivals) and the number of concurrently open tickets against the M/G/∞ prediction. The Poisson plan is evaluated twice: against a two-sided local mean over about 40 surrounding work days (in-sample, not a feasible forecast) and causally, with the trailing 20-work-day mean and a coefficient of variation estimated on earlier days only (rolling origin). Weekday effects and the work-week definition are estimated on the full sample, so only the level and the coefficient of variation are strictly causal.

Court cases. The Charlotin database (Charlotin, 2026; CC BY 4.0, accessed 6 October 2026) lists court decisions with AI-fabricated content. We use the decision date at day resolution for October 2025 to August 2026 and analyse weekday counts with the same stratified and local-mean estimators as for tickets. September 2026 (95 cases) and one case in October 2026 are excluded because of reporting lag.

WildChat. The WildChat-1M corpus (Zhao et al., 2024; ODC-BY; allenai/WildChat-1M on Hugging Face) holds 837,989 conversations with ChatGPT models, collected through a free research chatbot from 9 April 2023 to 29 April 2024 (387 days, three of them without a

conversation). We use only the timestamp of each conversation, which the corpus defines as that of its last turn, as the arrival time of one request. Daily counts are adjusted for the weekday effect, and the coefficient of variation of the daily rate is estimated as for tickets, against a leave-one-out local mean over about eight weeks, once for all days and once without days below 35% of the 41-day rolling median (five days). Within-hour variance-to-mean ratios are computed for the 30 busiest days at bin widths of 10 s, 1 min and 10 min. The stress test replays the measured daily multipliers on event streams with means of 10 and 120 per day (400 replications), and evaluates the Poisson plan in-sample against about 40 surrounding days and causally against the trailing 20 days.

Appendix C. Supplementary Tables

Table C1. Resolution and handling times of tickets (hours)

Stream / measure	n	Median	Mean	90th pct	99th pct	CV	Best fit (AIC) / Weibull shape
Software company: creation to resolution	22,016	167.8	698.8	1487	8806	3.0	weibull / 0.51
Software company: time in "in progress"	19,662	2.2	97.3	220	1532	3.6	weibull / 0.30
ServiceNow: opened to resolved	23,323	22.2	178.5	382	2849	3.0	weibull / 0.36
ServiceNow: Critical/High only	496	49.7	172.9	391	1086	2.0	weibull / 0.55
Italian help desk: first to last event	3,803	94.4	211.2	627	1008	1.3	weibull / 0.50

Calendar hours including waiting states (customer, vendor, queue); the in-progress measure counts only time in that workflow state. Exponential fits are rejected in every case (KS $p < 10^{-30}$).

Appendix D. Reproducibility and Data Availability

All code, per-query model outputs, guardrail decisions and analysis scripts are provided in a public repository [repository URL to be inserted]. The experiments run in the order collect (run_experiment.py collect), re-time with the extractive verifier (retime), analyse (run_experiment.py analyze, run_all_analyses.sh), summarise (summarize.py), compute the economic tables (econ_tables.py) and build the figures, tables and manuscript (make_figures.py,

paper_tables.py, paper/econ/build.js). Every number in the text and tables is generated from the resulting JSON files. The held-out replication of Section 5.7 (configuration fixed before the run, code, per-question outputs and analysis scripts) is in the same repository. The tables of Section 3 can be reproduced and checked with check_tables.py and econ_dependence.py, the replication of Section 5.7 with run_heldout.py, judge_api.py, analyze_heldout.py and cost_heldout.py, and the WildChat analysis with wildchat_arrivals.py. The public datasets are not redistributed; they are available from their original sources under the licences stated in Section 4.5 and Appendix B. Generator and verifier checkpoints are the public Hugging Face releases named in Section 4. All simulations and bootstrap draws are seeded. [Editorial note, delete before submission: the scripts named in the second sentence of this appendix belong to the original pipeline and were not among the files used to prepare this revision; please confirm that they are in the repository.]